Министерство образования Республики Беларусь

Учреждение образования
БЕЛОРУССКИЙ ГОСУДАРСТВЕННЫЙ УНИВЕРСИТЕТ
ИНФОРМАТИКИ И РАДИОЭЛЕКТРОНИКИ

Факультет информационных технологий и управления

Кафедра высшей математики

Дисциплина: Численные методы

Отчёт по лабораторной работе №4

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Вариант 1, подгруппа 2

In[22]:=
```
(*
Вариант 1:
  f(x) = 2^x, [-2,1]
*)
(*
   1. Постройте интерполяционные многочлены степени n для функции f(x),
   заданной в равноотстоящих точках отрезка[a,b] (согласно номера вашего варианта),
   и исследуйте зависимость погрешности интерполирования от
степени полинома n (n=1,2,3,4,5 и 10) (или, что равносильно,
      от расстояния между узлами h=(b-a)/n).Для этого:
     a) создайте таблицу значений заданной функции в равноотстоящих точках отрезка
  xj = a+(b-a)/n*j, j=0,1,...,n и выведите в виде списка и таблицы;
```

```
a = -2;
b = 1;
f[x_] := 2^x;
n = 1;
j = 0;
xs1 = {};
fxs1 = {};
While[j ≤ n,
  xj = a + (b - a) / n * j;
  xs1 = Append[xs1, xj];
  fxs1 = Append[fxs1, f[xj]];
  j = j + 1;
 ];
js = Range[0, n];
data = Transpose[{js, xs1, fxs1}];

TableForm[data, TableHeadings → {None, {"j", "xj", "f(xj)"}}]
```

Out[91]//TableForm=

j	xj	f(xj)
0	-2	$\frac{1}{4}$
1	1	2

In[93]:=
```
n = 2;
j = 0;
xs2 = {};
fxs2 = {};
While[j ≤ n,
  xj = a + (b - a) / n * j;
  xs2 = Append[xs2, xj];
  fxs2 = Append[fxs2, f[xj]];
  j = j + 1;
 ];
js = Range[0, n];
data = Transpose[{js, xs2, fxs2}];

TableForm[data, TableHeadings → {None, {"j", "xj", "f(xj)"}}]
```

Out[100]//TableForm=

j	xj	f(xj)
0	-2	$\frac{1}{4}$
1	$-\frac{1}{2}$	$\frac{1}{\sqrt{2}}$
2	1	2

In[149]:=
```
n = 3;
j = 0;
xs3 = {};
fxs3 = {};
While[j ≤ n,
  xj = a + (b - a) / n * j;
  xs3 = Append[xs3, xj];
  fxs3 = Append[fxs3, f[xj]];
  j = j + 1;
 ];
js = Range[0, n];
data = Transpose[{js, xs3, fxs3}];

TableForm[data, TableHeadings → {None, {"j", "xj", "f(xj)"}}]
```

Out[156]//TableForm=

j	xj	f(xj)
0	-2	$\frac{1}{4}$
1	-1	$\frac{1}{2}$
2	0	1
3	1	2

In[109]:=
```
n = 4;
j = 0;
xs4 = {};
fxs4 = {};
While[j ≤ n,
  xj = a + (b - a) / n * j;
  xs4 = Append[xs4, xj];
  fxs4 = Append[fxs4, f[xj]];
  j = j + 1;
 ];
js = Range[0, n];
data = Transpose[{js, xs4, fxs4}];

TableForm[data, TableHeadings → {None, {"j", "xj", "f(xj)"}}]
```

Out[116]//TableForm=

j	xj	f(xj)
0	-2	$\frac{1}{4}$
1	$-\frac{5}{4}$	$\frac{1}{2\ 2^{1/4}}$
2	$-\frac{1}{2}$	$\frac{1}{\sqrt{2}}$
3	$\frac{1}{4}$	$2^{1/4}$
4	1	2

In[117]:=
```
n = 5;
j = 0;
xs5 = {};
fxs5 = {};
While[j ≤ n,
  xj = a + (b - a) / n * j;
  xs5 = Append[xs5, xj];
  fxs5 = Append[fxs5, f[xj]];
  j = j + 1;
 ];
js = Range[0, n];
data = Transpose[{js, xs5, fxs5}];

TableForm[data, TableHeadings → {None, {"j", "xj", "f(xj)"}}]
```

Out[124]//TableForm=

j	xj	f(xj)
0	-2	$\frac{1}{4}$
1	$-\frac{7}{5}$	$\frac{1}{2\times 2^{2/5}}$
2	$-\frac{4}{5}$	$\frac{1}{2^{4/5}}$
3	$-\frac{1}{5}$	$\frac{1}{2^{1/5}}$
4	$\frac{2}{5}$	$2^{2/5}$
5	1	2

In[141]:=
```
n = 10;
j = 0;
xs10 = {};
fxs10 = {};
While[j ≤ n,
  xj = a + (b - a) / n * j;
  xs10 = Append[xs10, xj];
  fxs10 = Append[fxs10, f[xj]];
  j = j + 1;
 ];
js = Range[0, n];
data = Transpose[{js, xs10, fxs10}];

TableForm[data, TableHeadings → {None, {"j", "xj", "f(xj)"}}]
```

Out[148]//TableForm=

j	xj	f(xj)
0	-2	$\frac{1}{4}$
1	$-\frac{17}{10}$	$\frac{1}{2\cdot 2^{7/10}}$
2	$-\frac{7}{5}$	$\frac{1}{2\cdot 2^{2/5}}$
3	$-\frac{11}{10}$	$\frac{1}{2\cdot 2^{1/10}}$
4	$-\frac{4}{5}$	$\frac{1}{2^{4/5}}$
5	$-\frac{1}{2}$	$\frac{1}{\sqrt{2}}$
6	$-\frac{1}{5}$	$\frac{1}{2^{1/5}}$
7	$\frac{1}{10}$	$2^{1/10}$
8	$\frac{2}{5}$	$2^{2/5}$
9	$\frac{7}{10}$	$2^{7/10}$
10	1	2

In[74]:=
```
(*
  б) найдите интерполяционный многочлен N(X) n с помощью встроенной функции
 InterpolatingPolinomial и выведите его,упростите вид полученного
  многочлена при помощи функции Expand или Collect;
*)
data1 = Transpose[{xs1, fxs1}]
P1 = InterpolatingPolynomial[data1, x]
```

Out[74]= $\left\{\left\{-2, \frac{1}{4}\right\}, \{1, 2\}\right\}$

Out[75]= $\frac{1}{4} + \frac{7\,(2 + x)}{12}$

In[76]:=
```
data2 = Transpose[{xs2, fxs2}]
P2 = Expand[InterpolatingPolynomial[data2, x]]
```

Out[76]= $\left\{\left\{-2, \frac{1}{4}\right\}, \left\{-\frac{1}{2}, \frac{1}{\sqrt{2}}\right\}, \{1, 2\}\right\}$

Out[77]= $\frac{5}{12} + \frac{4\sqrt{2}}{9} + \frac{13\,x}{12} - \frac{2\sqrt{2}\,x}{9} + \frac{x^2}{2} - \frac{2\sqrt{2}\,x^2}{9}$

In[78]:=
```
data3 = Transpose[{xs3, fxs3}]
P3 = Expand[InterpolatingPolynomial[data3, x]]
```

Out[78]= $\left\{\left\{-2, \frac{1}{4}\right\}, \left\{-1, \frac{1}{2}\right\}, \{0, 1\}, \{1, 2\}\right\}$

Out[79]= $1 + \frac{17\,x}{24} + \frac{x^2}{4} + \frac{x^3}{24}$

In[80]:=
```
data4 = Transpose[{xs4, fxs4}]
P4 = Expand[InterpolatingPolynomial[data4, x]]
```

Out[80]= $\left\{\left\{-2, \frac{1}{4}\right\}, \left\{-\frac{5}{4}, \frac{1}{2\times 2^{1/4}}\right\}, \left\{-\frac{1}{2}, \frac{1}{\sqrt{2}}\right\}, \left\{\frac{1}{4}, 2^{1/4}\right\}, \{1, 2\}\right\}$

Out[81]= $-\frac{25}{324} + \frac{160\times 2^{1/4}}{243} + \frac{20\sqrt{2}}{81} - \frac{8\times 2^{3/4}}{243} + \frac{5\,x}{108} + \frac{368}{243}\times 2^{1/4}\,x - \frac{74\sqrt{2}\,x}{81} + \frac{20}{243}\times 2^{3/4}\,x + \frac{43\,x^2}{54} - \frac{16}{81}\times 2^{1/4}\,x^2 -$
$\frac{14\sqrt{2}\,x^2}{27} + \frac{20}{81}\times 2^{3/4}\,x^2 + \frac{76\,x^3}{81} - \frac{352}{243}\times 2^{1/4}\,x^3 + \frac{64\sqrt{2}\,x^3}{81} - \frac{40}{243}\times 2^{3/4}\,x^3 + \frac{8\,x^4}{27} - \frac{128}{243}\times 2^{1/4}\,x^4 + \frac{32\sqrt{2}\,x^4}{81} - \frac{32}{243}\times 2^{3/4}\,x^4$

In[82]:=
```
data5 = Transpose[{xs5, fxs5}]
P5 = Expand[InterpolatingPolynomial[data5, x]]
```

Out[82]= $\left\{\left\{-2, \frac{1}{4}\right\}, \left\{-\frac{7}{5}, \frac{1}{2\times 2^{2/5}}\right\}, \left\{-\frac{4}{5}, \frac{1}{2^{4/5}}\right\}, \left\{-\frac{1}{5}, \frac{1}{2^{1/5}}\right\}, \left\{\frac{2}{5}, 2^{2/5}\right\}, \{1, 2\}\right\}$

Out[83]= $-\frac{119}{2916} - \frac{175}{729\times 2^{4/5}} + \frac{25}{729\times 2^{2/5}} + \frac{175\times 2^{2/5}}{729} + \frac{350\times 2^{4/5}}{729} - \frac{2321\,x}{11\,664} - \frac{475\,x}{729\times 2^{4/5}} + \frac{4525\,x}{1458\times 2^{3/5}} + \frac{325\,x}{2916\times 2^{2/5}} - \frac{725\,x}{729\times 2^{1/5}} + \frac{4625\,x^2}{23\,328} + \frac{8975\,x^2}{2916\times 2^{4/5}} + \frac{9425\,x^2}{2916\times 2^{3/5}} - \frac{1175\,x^2}{2916\times 2^{2/5}} - \frac{2975\,x^2}{729\times 2^{1/5}} + \frac{23\,875\,x^3}{23\,328} +$
$\frac{3625\,x^3}{2916\times 2^{4/5}} - \frac{6125\,x^3}{2916\times 2^{3/5}} - \frac{5125\,x^3}{11\,664\times 2^{2/5}} + \frac{125\,x^3}{2916\times 2^{1/5}} + \frac{19\,375\,x^4}{23\,328} - \frac{6875\,x^4}{2916\times 2^{4/5}} - \frac{10\,625\,x^4}{2916\times 2^{3/5}} + \frac{625\,x^4}{1458\times 2^{2/5}} + \frac{4375\,x^4}{1458\times 2^{1/5}} + \frac{4375\,x^5}{23\,328} - \frac{3125\,x^5}{2916\times 2^{4/5}} - \frac{3125\,x^5}{2916\times 2^{3/5}} + \frac{3125\,x^5}{11\,664\times 2^{2/5}} + \frac{3125\,x^5}{2916\times 2^{1/5}}$

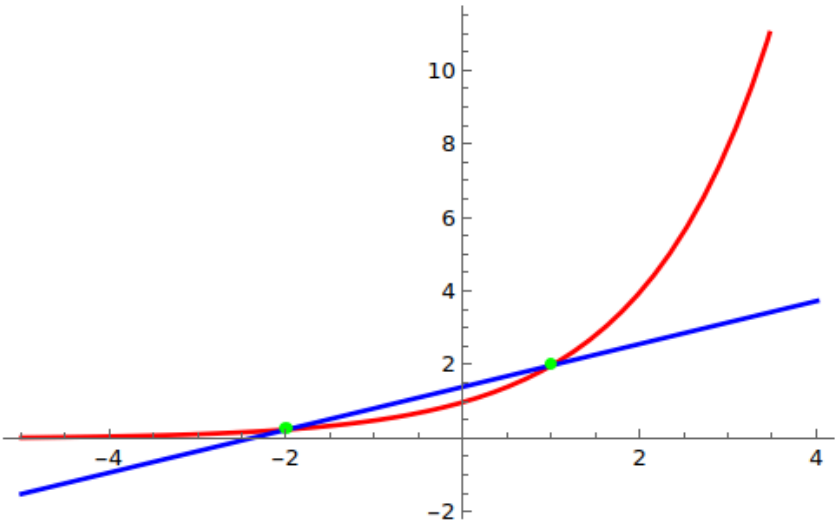
In[84]:=
```
data10 = Transpose[{xs10, fxs10}]
P10 = Expand[InterpolatingPolynomial[data10, x]]
```

Out[84]=
$$\left\{\left\{-2,\frac{1}{4}\right\},\left\{-\frac{1}{10},\frac{1}{2\times 2^{7/10}}\right\},\left\{-\frac{1}{5},\frac{1}{2\times 2^{2/5}}\right\},\left\{-\frac{11}{10},\frac{1}{2\times 2^{1/10}}\right\},\left\{\frac{4}{5},\frac{1}{2^{4/5}}\right\},\left\{-\frac{1}{2},\frac{1}{\sqrt{2}}\right\},\left\{-\frac{1}{5},\frac{1}{2^{1/5}}\right\},\left\{\frac{1}{10},2^{1/10}\right\},\left\{\frac{2}{5},2^{2/5}\right\},\left\{\frac{1}{10},2^{7/10}\right\},\{1,2\}\right\}$$

Out[85]=
$$-\frac{6545}{6\,377\,292}+\frac{229\,075}{1\,594\,323\times 2^{4/5}}+\frac{4675}{531\,441\times 2^{2/5}}+\frac{1\,047\,200\times 2^{1/10}}{1\,594\,323}-\frac{3850\times 2^{3/10}}{4\,782\,969}-\frac{32\,725\times 2^{2/5}}{531\,441}-\frac{73\,304\,\sqrt{2}}{531\,441}+\frac{37\,400\times 2^{7/10}}{4\,782\,969}+\frac{458\,150\times 2^{4/5}}{1\,594\,323}-\frac{23\,800\times 2^{9/10}}{1\,594\,323}+\frac{25\,019\,x}{8\,503\,056}-\frac{1\,195\,075\,x}{1\,594\,323\times 2^{4/5}}+\frac{35\,075\,x}{4\,782\,969\times 2^{7/10}}+$$
$$\frac{192\,025\,x}{1\,062\,882\times 2^{3/5}}-\frac{612\,775\,x}{14\,880\,348\times 2^{2/5}}-\frac{8\,216\,425\,x}{1\,594\,323\times 2^{1/5}}+\frac{6\,317\,800\times 2^{1/10}\,x}{1\,594\,323}+\frac{437\,402\,\sqrt{2}\,x}{531\,441}-\frac{664\,550\times 2^{7/10}\,x}{33\,480\,783}+\frac{116\,050\times 2^{9/10}\,x}{1\,594\,323}+\frac{2\,701\,589\,x^2}{39\,680\,928}-\frac{5\,752\,675\,x^2}{708\,588\times 2^{4/5}}+\frac{13\,579\,975\,x^2}{1\,653\,372\times 2^{3/5}}-$$
$$\frac{868\,025\,x^2}{1\,653\,372\times 2^{2/5}}-\frac{1\,851\,925\,x^2}{177\,147\times 2^{1/5}}-\frac{2\,194\,400\times 2^{1/10}\,x^2}{1\,240\,029}+\frac{180\,700\times 2^{3/10}\,x^2}{3\,720\,087}+\frac{416\,128\,\sqrt{2}\,x^2}{59\,049}-\frac{1\,938\,800\times 2^{7/10}\,x^2}{3\,720\,087}+\frac{1\,084\,600\times 2^{9/10}\,x^2}{1\,240\,029}+\frac{11\,725\,375\,x^3}{119\,042\,784}+\frac{14\,377\,775\,x^3}{2\,125\,764\times 2^{4/5}}-\frac{858\,175\,x^3}{44\,641\,044\times 2^{7/10}}+$$
$$\frac{116\,239\,325\,x^3}{4\,960\,116\times 2^{3/5}}-\frac{3\,842\,375\,x^3}{177\,147\,\sqrt{2}}+\frac{3\,328\,925\,x^3}{19\,840\,464\times 2^{2/5}}-\frac{20\,729\,575\,x^3}{11\,160\,261\times 2^{3/10}}+\frac{82\,336\,475\,x^3}{2\,125\,764\times 2^{1/5}}-\frac{3\,299\,725\,x^3}{3\,720\,087\times 2^{1/10}}-\frac{17\,627\,150\times 2^{1/10}\,x^3}{531\,441}-\frac{183\,875\,x^4}{489\,888}+\frac{122\,836\,375\,x^4}{2\,125\,764\times 2^{4/5}}-\frac{32\,626\,625\,x^4}{44\,641\,044\times 2^{7/10}}+$$
$$\frac{8\,981\,125\,x^4}{4\,960\,116\times 2^{3/5}}-\frac{14\,844\,625\,x^4}{177\,147\,\sqrt{2}}+\frac{39\,042\,125\,x^4}{9\,920\,232\times 2^{2/5}}+\frac{50\,848\,375\,x^4}{11\,160\,261\times 2^{3/10}}-\frac{48\,203\,375\,x^4}{3\,720\,087\times 2^{1/10}}-\frac{107\,431\,750\times 2^{1/10}\,x^4}{3\,720\,087}+\frac{18\,944\,000\times 2^{4/5}\,x^4}{531\,441}-\frac{5\,453\,125\,x^5}{5\,668\,704}+\frac{2\,354\,375\,x^5}{708\,588\times 2^{4/5}}-\frac{885\,625\,x^5}{2\,125\,764\times 2^{7/10}}-$$
$$\frac{14\,226\,875\,x^5}{236\,196\times 2^{3/5}}+\frac{1\,330\,625\,x^5}{59\,049\,\sqrt{2}}+\frac{1\,714\,375\,x^5}{944\,784\times 2^{2/5}}+\frac{7\,829\,375\,x^5}{531\,441\times 2^{3/10}}-\frac{38\,441\,875\,x^5}{708\,588\times 2^{1/5}}-\frac{694\,375\,x^5}{177\,147\times 2^{1/10}}+\frac{10\,026\,250\times 2^{1/10}\,x^5}{177\,147}-\frac{2\,088\,125\,x^6}{8\,503\,056}-\frac{115\,553\,125\,x^6}{1\,062\,882\times 2^{4/5}}+\frac{9\,034\,375\,x^6}{6\,377\,292\times 2^{7/10}}-\frac{21\,596\,875\,x^6}{354\,294\times 2^{3/5}}+$$
$$\frac{27\,580\,625\,x^6}{177\,147\,\sqrt{2}}-\frac{10\,909\,375\,x^6}{1\,417\,176\times 2^{2/5}}+\frac{15\,409\,375\,x^6}{1\,594\,323\times 2^{3/10}}-\frac{156\,240\,625\,x^6}{1\,062\,882\times 2^{1/5}}+\frac{13\,440\,625\,x^6}{531\,441\times 2^{1/10}}+\frac{46\,756\,250\times 2^{1/10}\,x^6}{531\,441}+\frac{859\,375\,x^7}{708\,588}-\frac{15\,453\,125\,x^7}{2\,480\,058\times 2^{2/5}}+\frac{2\,125\,000\times 2^{1/10}\,x^7}{531\,441}-\frac{14\,796\,875\times 2^{1/5}\,x^7}{531\,441}+$$
$$\frac{1\,046\,875\times 2^{3/10}\,x^7}{1\,594\,323}+\frac{1\,046\,875\times 2^{2/5}\,x^7}{177\,147}+\frac{5\,000\,000\,\sqrt{2}\,x^7}{177\,147}-\frac{39\,312\,500\times 2^{7/10}\,x^7}{11\,160\,261}-\frac{8\,328\,125\times 2^{4/5}\,x^7}{531\,441}+\frac{4\,468\,750\times 2^{9/10}\,x^7}{531\,441}+\frac{1\,796\,875\,x^8}{1\,240\,029}+\frac{1\,328\,125\,x^8}{413\,343\times 2^{2/5}}-\frac{65\,000\,000\times 2^{1/10}\,x^8}{1\,240\,029}+$$
$$\frac{4\,375\,000\times 2^{1/5}\,x^8}{177\,147}-\frac{1\,015\,625\times 2^{3/10}\,x^8}{3\,720\,087}+\frac{9\,687\,500\times 2^{2/5}\,x^8}{413\,343}-\frac{2\,187\,500\,\sqrt{2}\,x^8}{59\,049}-\frac{22\,812\,500\times 2^{7/10}\,x^8}{3\,720\,087}+\frac{6\,718\,750\times 2^{4/5}\,x^8}{177\,147}-\frac{6\,875\,000\times 2^{9/10}\,x^8}{1\,240\,029}+\frac{14\,453\,125\,x^9}{22\,320\,522}-\frac{16\,015\,625\,x^9}{3\,720\,087\times 2^{2/5}}-$$
$$\frac{50\,000\,000\times 2^{1/10}\,x^9}{1\,594\,323}+\frac{36\,718\,750\times 2^{1/5}\,x^9}{1\,594\,323}-\frac{14\,843\,750\times 2^{3/10}\,x^9}{33\,480\,783}+\frac{46\,093\,750\times 2^{2/5}\,x^9}{3\,720\,087}-\frac{15\,625\,000\,\sqrt{2}\,x^9}{531\,441}-\frac{96\,875\,000\times 2^{7/10}\,x^9}{33\,480\,783}+\frac{41\,406\,250\times 2^{4/5}\,x^9}{1\,594\,323}-\frac{68\,750\,000\times 2^{9/10}\,x^9}{11\,160\,261}+\frac{390\,625\,x^{10}}{3\,720\,087}-$$
$$\frac{62\,500\,000\times 2^{1/10}\,x^{10}}{11\,160\,261}+\frac{7\,812\,500\times 2^{1/5}\,x^{10}}{1\,594\,323}-\frac{3\,906\,250\times 2^{3/10}\,x^{10}}{33\,480\,783}+\frac{7\,812\,500\times 2^{2/5}\,x^{10}}{3\,720\,087}-\frac{3\,125\,000\,\sqrt{2}\,x^{10}}{531\,441}+\frac{1\,953\,125\times 2^{3/5}\,x^{10}}{3\,720\,087}-\frac{15\,625\,000\times 2^{7/10}\,x^{10}}{33\,480\,783}+\frac{7\,812\,500\times 2^{4/5}\,x^{10}}{1\,594\,323}-\frac{15\,625\,000\times 2^{9/10}\,x^{10}}{11\,160\,261}$$

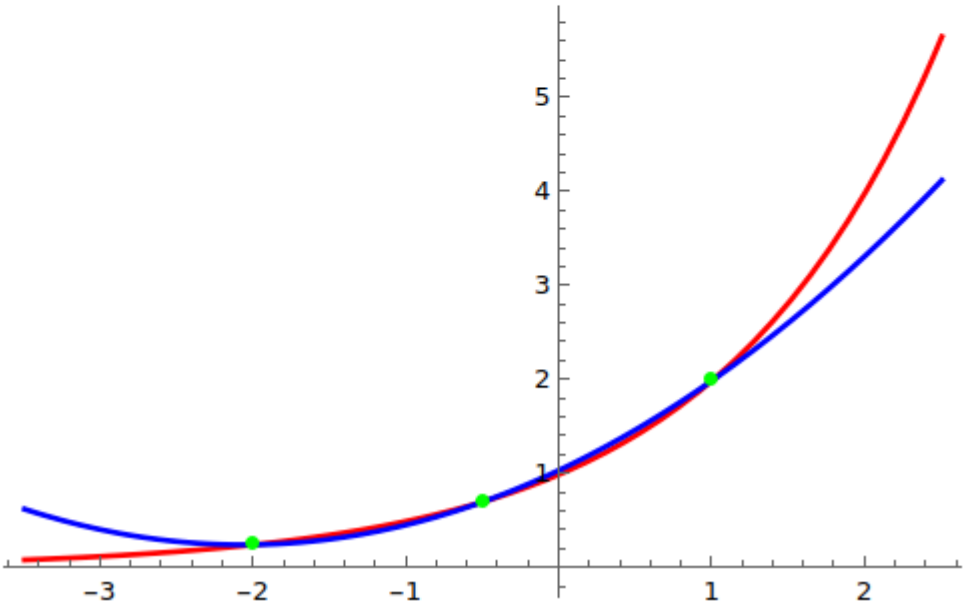
In[181]:=
```
(*
    в) выведите графики функции f(x),узлов интерполирования и
    интерполяционного многочлена N(x) n на отрезке[a-h,b+h];
*)
n = 1;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P1, {x, xmin, xmax}, PlotStyle → Blue];
p31 = ListPlot[data1, PlotStyle → Green];
Show[p1, p2, p31, PlotRange → All]
```

Out[188]=



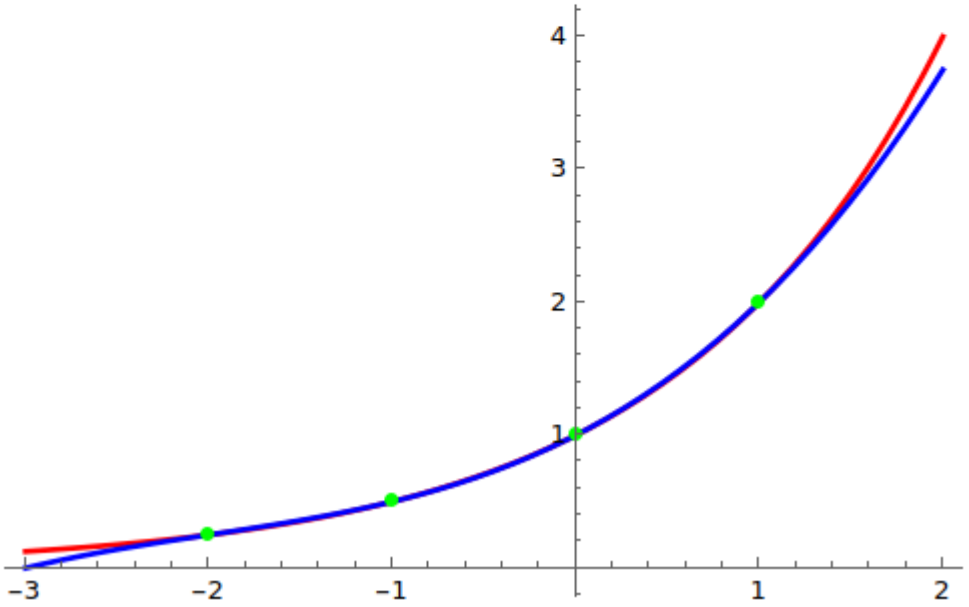
In[197]:=
```
n = 2;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P2, {x, xmin, xmax}, PlotStyle → Blue];
p32 = ListPlot[data2, PlotStyle → Green];
Show[p1, p2, p32, PlotRange → All]
```

Out[204]=



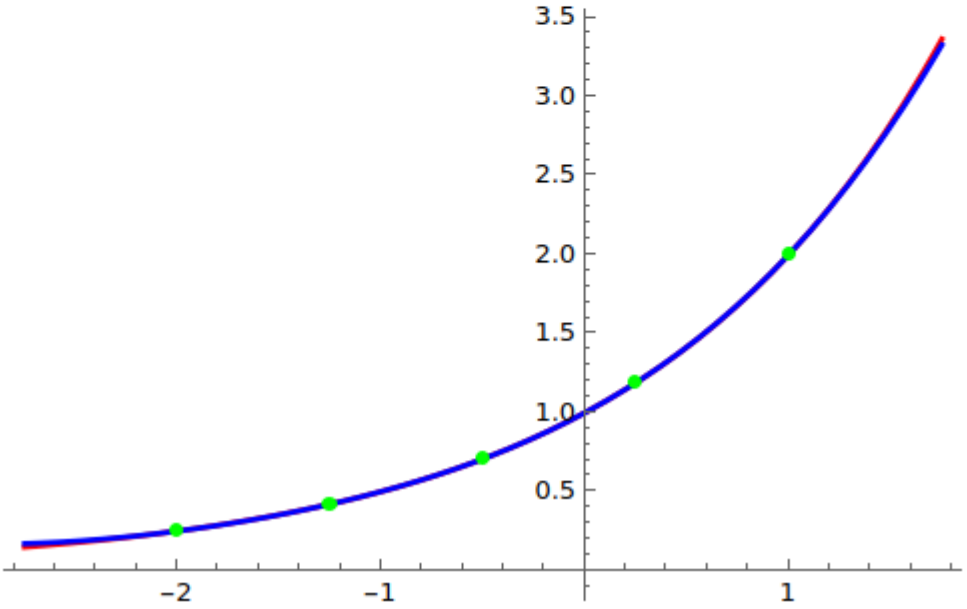
In[205]:=
```
n = 3;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P3, {x, xmin, xmax}, PlotStyle → Blue];
p33 = ListPlot[data3, PlotStyle → Green];
Show[p1, p2, p33, PlotRange → All]
```

Out[212]=



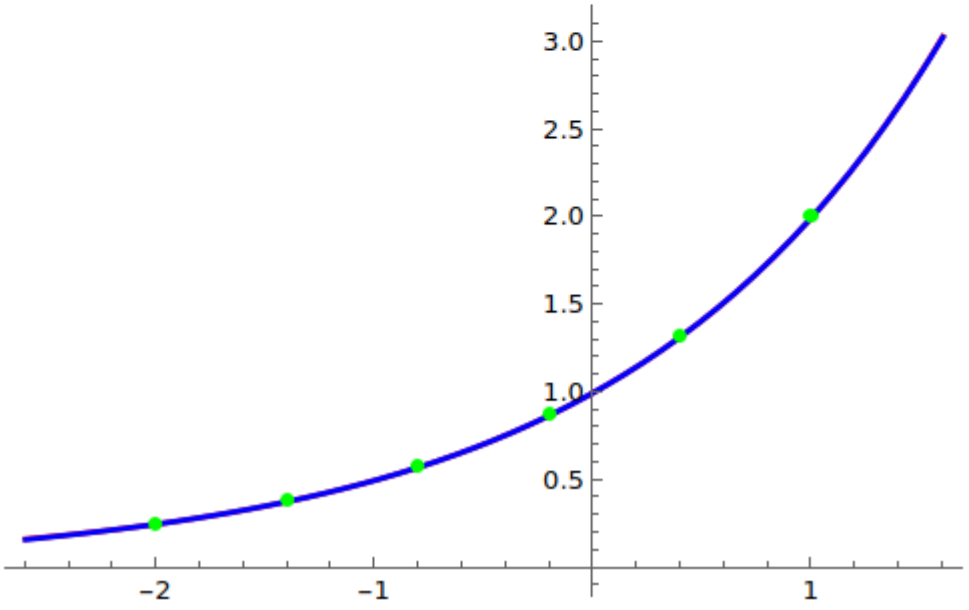
In[213]:=
```
n = 4;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P4, {x, xmin, xmax}, PlotStyle → Blue];
p34 = ListPlot[data4, PlotStyle → Green];
Show[p1, p2, p34, PlotRange → All]
```

Out[220]=



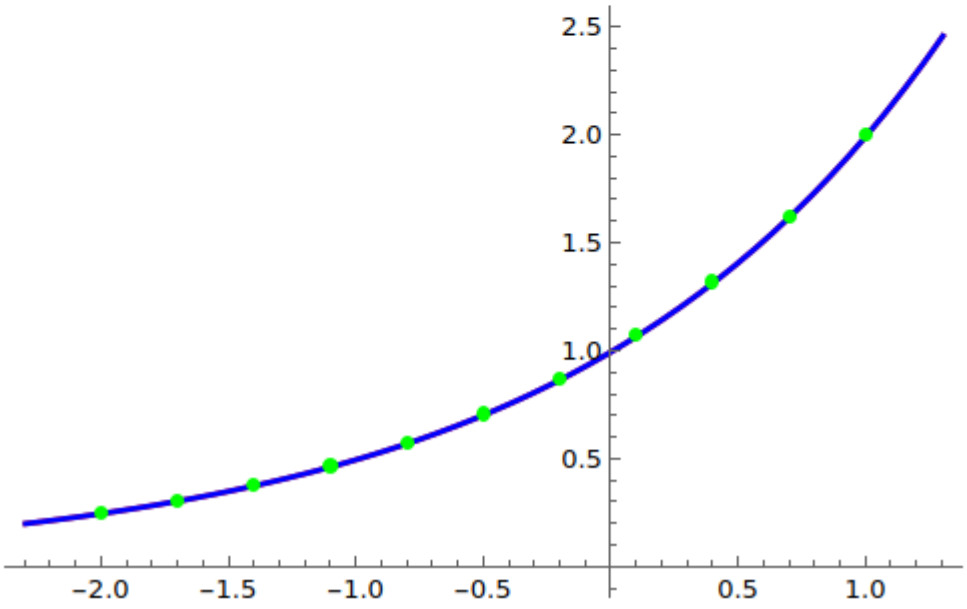
In[221]:=
```
n = 5;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P5, {x, xmin, xmax}, PlotStyle → Blue];
p35 = ListPlot[data5, PlotStyle → Green];
Show[p1, p2, p35, PlotRange → All]
```

Out[228]=



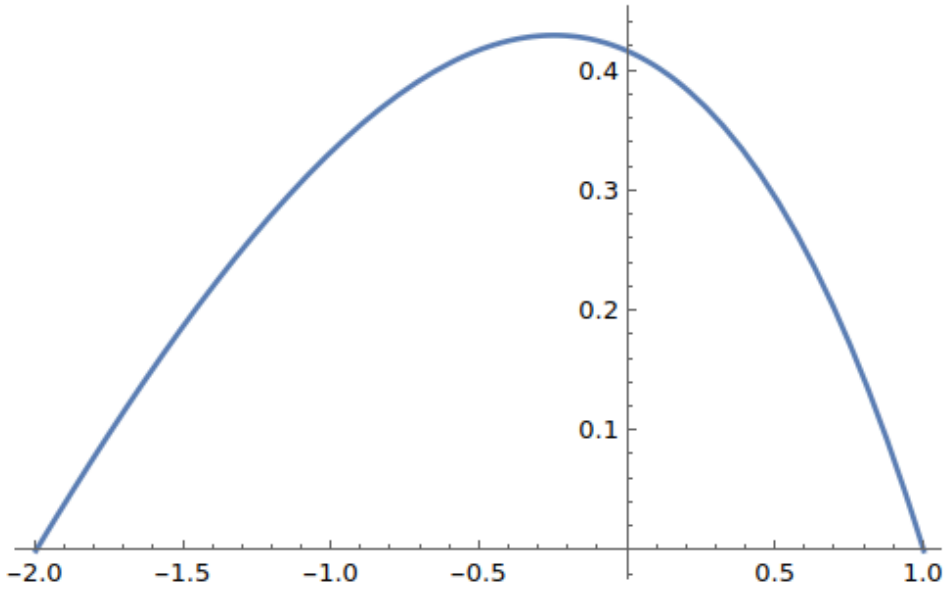
In[229]:=
```
n = 10;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P10, {x, xmin, xmax}, PlotStyle → Blue];
p310 = ListPlot[data10, PlotStyle → Green];
Show[p1, p2, p310, PlotRange → All]
```

Out[236]=



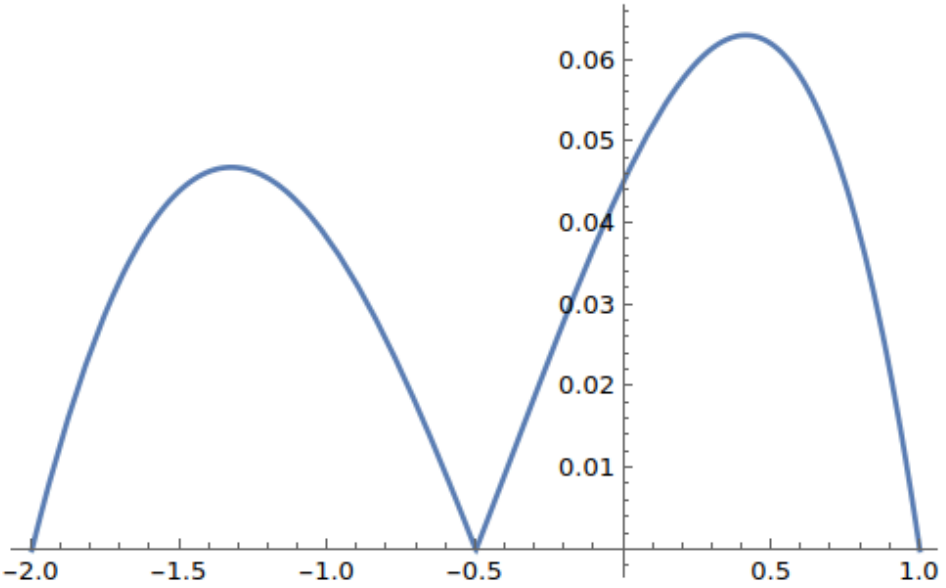
In[241]:=
```
Rn1 = Plot[Abs[f[x] - P1], {x, a, b}, PlotRange → All]
```

Out[241]=



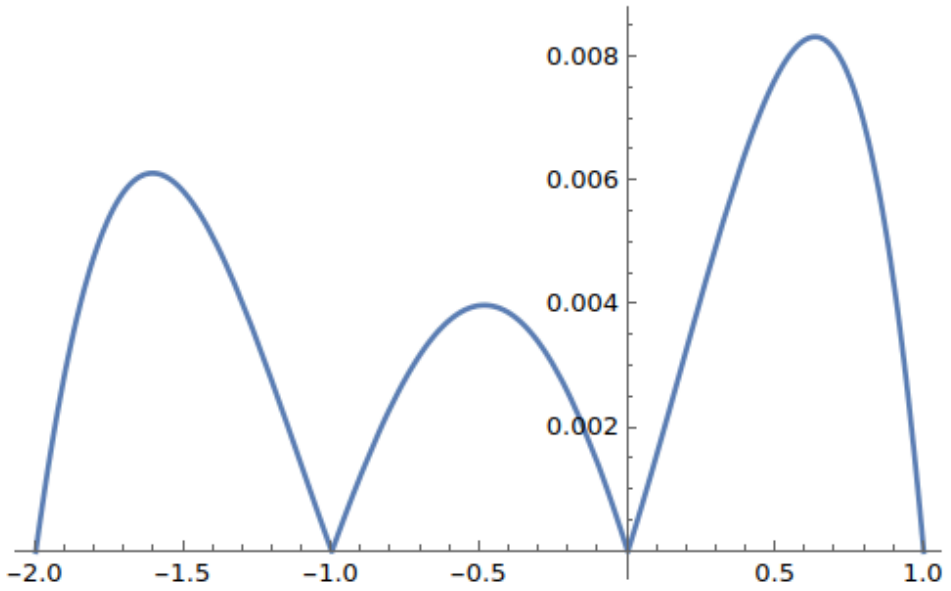
In[242]:=
```
Rn2 = Plot[Abs[f[x] - P2], {x, a, b}, PlotRange → All]
```

Out[242]=



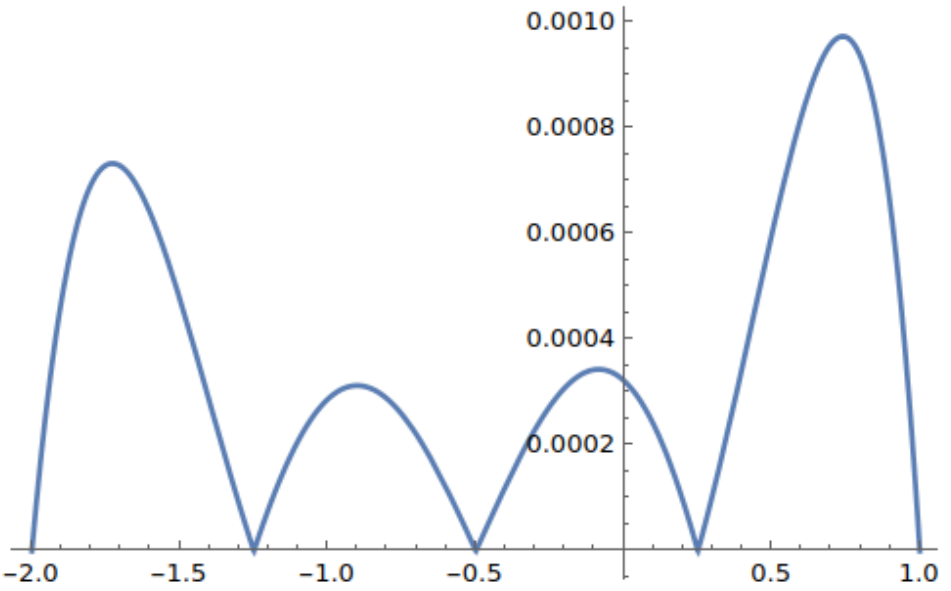
In[245]:=
```
Rn3 = Plot[Abs[f[x] - P3], {x, a, b}, PlotRange → All]
```

Out[245]=



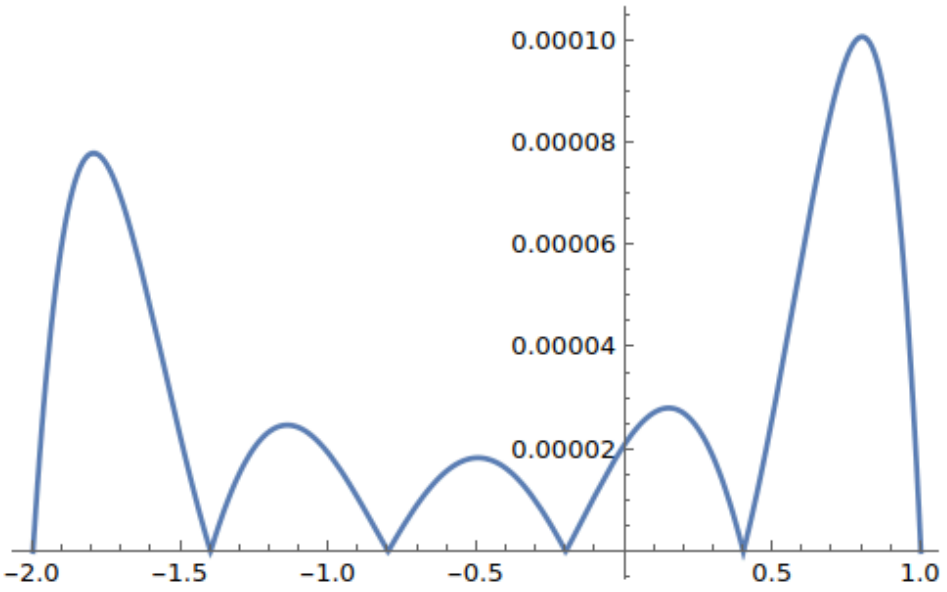
In[246]:=
```
Rn4 = Plot[Abs[f[x] - P4], {x, a, b}, PlotRange → All]
```

Out[246]=



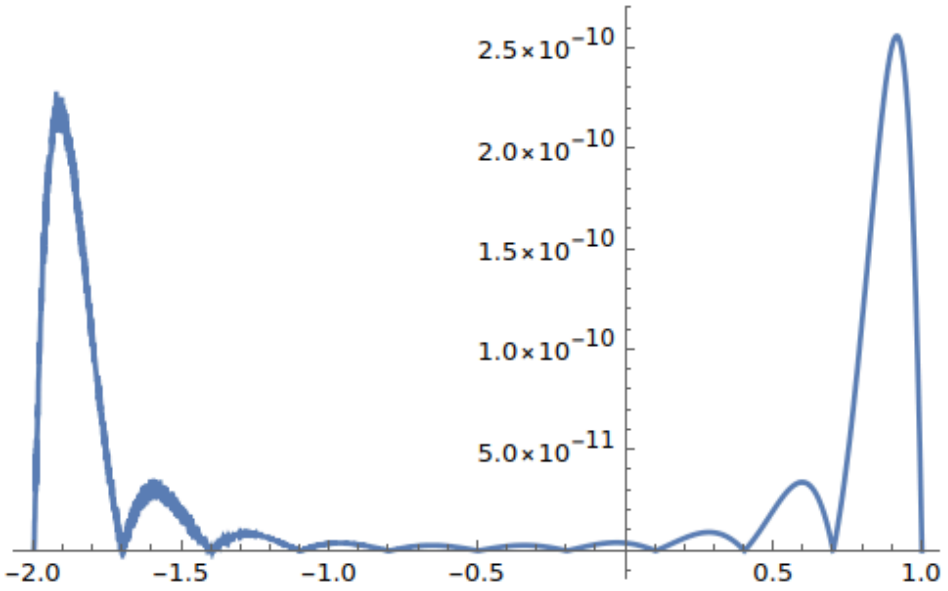
In[247]:= `Rn5 = Plot[Abs[f[x] - P5], {x, a, b}, PlotRange → All]`

Out[247]=



In[248]:= `Rn10 = Plot[Abs[f[x] - P10], {x, a, b}, PlotRange → All]`

Out[248]=



In[252]:= FindMaximum[{Abs[f[x] - P1], a ≤ x ≤ b}, x]

Out[252]= {0.429937, {x → -0.248841}}

In[253]:= FindMaximum[{Abs[f[x] - P2], a ≤ x ≤ b}, x]

Out[253]= {0.0630579, {x → 0.41145}}

In[254]:= FindMaximum[{Abs[f[x] - P3], a ≤ x ≤ b}, x]

Out[254]= {0.00832901, {x → 0.632954}}

In[255]:= FindMaximum[{Abs[f[x] - P4], a ≤ x ≤ b}, x]

Out[255]= {0.000974132, {x → 0.739772}}

In[256]:= FindMaximum[{Abs[f[x] - P5], a ≤ x ≤ b}, x]

Out[256]= {0.000100931, {x → 0.801253}}

In[257]:= FindMaximum[{Abs[f[x] - P10], a ≤ x ≤ b}, x]

Out[257]= $\left\{2.43027 \times 10^{-11}, \{x \to 0.517105\}\right\}$

2. Повторите п.1 для неравноотстоящих узлов на отрезке[a,b]-узлы расположить пропорционально нулям многочленов Чебышева:
xk = (a+b)/2+(b-a)/2 * cos((2k+1)*pi/(2(n+1))), k=0,1,...,n
*)

```
n = 1;
k = 0;
xs1 = {};
fxs1 = {};
While[k ≤ n,
  xk = (a + b) / 2 + (b - a) / 2 * Cos[(2 k + 1) * Pi / (2 (n + 1))];
  xs1 = Append[xs1, xk];
  fxs1 = Append[fxs1, f[xk]];
  k = k + 1;
 ];
ks = Range[0, n];
data = Transpose[{ks, xs1, fxs1}];

TableForm[data, TableHeadings → {None, {"j", "xk", "f(xk)"}}]
```

ut[16]//TableForm=

j	xk	f(xk)
0	$-\frac{1}{2}+\frac{3}{2\sqrt{2}}$	$2^{-\frac{1}{2}+\frac{3}{2\sqrt{2}}}$
1	$-\frac{1}{2}-\frac{3}{2\sqrt{2}}$	$2^{-\frac{1}{2}-\frac{3}{2\sqrt{2}}}$

In[17]:=
```
n = 2;
k = 0;
xs2 = {};
fxs2 = {};
While[k ≤ n,
  xk = (a + b) / 2 + (b - a) / 2 * Cos[(2 k + 1) * Pi / (2 (n + 1))];
  xs2 = Append[xs2, xk];
  fxs2 = Append[fxs2, f[xk]];
  k = k + 1;
 ];
ks = Range[0, n];
data = Transpose[{ks, xs2, fxs2}];

TableForm[data, TableHeadings → {None, {"j", "xk", "f(xk)"}}]
```

Out[24]//TableForm=

j	xk	f(xk)
0	$-\frac{1}{2}+\frac{3\sqrt{3}}{4}$	$2^{-\frac{1}{2}+\frac{3\sqrt{3}}{4}}$
1	$-\frac{1}{2}$	$\frac{1}{\sqrt{2}}$
2	$-\frac{1}{2}-\frac{3\sqrt{3}}{4}$	$2^{-\frac{1}{2}-\frac{3\sqrt{3}}{4}}$

In[25]:=
```
n = 3;
k = 0;
xs3 = {};
fxs3 = {};
While[k ≤ n,
  xk = (a + b) / 2 + (b - a) / 2 * Cos[(2 k + 1) * Pi / (2 (n + 1))];
  xs3 = Append[xs3, xk];
  fxs3 = Append[fxs3, f[xk]];
  k = k + 1;
 ];
ks = Range[0, n];
data = Transpose[{ks, xs3, fxs3}];

TableForm[data, TableHeadings → {None, {"j", "xk", "f(xk)"}}]
```

ut[32]//TableForm=

j	xk	f(xk)
0	$-\frac{1}{2}+\frac{3}{2}\,\text{Cos}\left[\frac{\pi}{8}\right]$	$2^{-\frac{1}{2}+\frac{3}{2}\,\text{Cos}\left[\frac{\pi}{8}\right]}$
1	$-\frac{1}{2}+\frac{3}{2}\,\text{Sin}\left[\frac{\pi}{8}\right]$	$2^{-\frac{1}{2}+\frac{3}{2}\,\text{Sin}\left[\frac{\pi}{8}\right]}$
2	$-\frac{1}{2}-\frac{3}{2}\,\text{Sin}\left[\frac{\pi}{8}\right]$	$2^{-\frac{1}{2}-\frac{3}{2}\,\text{Sin}\left[\frac{\pi}{8}\right]}$
3	$-\frac{1}{2}-\frac{3}{2}\,\text{Cos}\left[\frac{\pi}{8}\right]$	$2^{-\frac{1}{2}-\frac{3}{2}\,\text{Cos}\left[\frac{\pi}{8}\right]}$

In[33]:=
```
n = 4;
k = 0;
xs4 = {};
fxs4 = {};
While[k ≤ n,
  xk = (a + b) / 2 + (b - a) / 2 * Cos[(2 k + 1) * Pi / (2 (n + 1))];
  xs4 = Append[xs4, xk];
  fxs4 = Append[fxs4, f[xk]];
  k = k + 1;
 ];
ks = Range[0, n];
data = Transpose[{ks, xs4, fxs4}];

TableForm[data, TableHeadings → {None, {"j", "xk", "f(xk)"}}]
```

Out[40]//TableForm=

j	xk	f(xk)
0	$-\frac{1}{2}+\frac{3}{2}\sqrt{\frac{5}{8}+\frac{\sqrt{5}}{8}}$	$2^{-\frac{1}{2}+\frac{3}{2}\sqrt{\frac{5}{8}+\frac{\sqrt{5}}{8}}}$
1	$-\frac{1}{2}+\frac{3}{2}\sqrt{\frac{5}{8}-\frac{\sqrt{5}}{8}}$	$2^{-\frac{1}{2}+\frac{3}{2}\sqrt{\frac{5}{8}-\frac{\sqrt{5}}{8}}}$
2	$-\frac{1}{2}$	$\frac{1}{\sqrt{2}}$
3	$-\frac{1}{2}-\frac{3}{2}\sqrt{\frac{5}{8}-\frac{\sqrt{5}}{8}}$	$2^{-\frac{1}{2}-\frac{3}{2}\sqrt{\frac{5}{8}-\frac{\sqrt{5}}{8}}}$
4	$-\frac{1}{2}-\frac{3}{2}\sqrt{\frac{5}{8}+\frac{\sqrt{5}}{8}}$	$2^{-\frac{1}{2}-\frac{3}{2}\sqrt{\frac{5}{8}+\frac{\sqrt{5}}{8}}}$

In[41]:=
```
n = 5;
k = 0;
xs5 = {};
fxs5 = {};
While[k ≤ n,
  xk = (a + b) / 2 + (b - a) / 2 * Cos[(2 k + 1) * Pi / (2 (n + 1))];
  xs5 = Append[xs5, xk];
  fxs5 = Append[fxs5, f[xk]];
  k = k + 1;
 ];
ks = Range[0, n];
data = Transpose[{ks, xs5, fxs5}];

TableForm[data, TableHeadings → {None, {"j", "xk", "f(xk)"}}]
```

Out[48]//TableForm=

j	xk	f(xk)
0	$-\frac{1}{2}+\frac{3\left(1+\sqrt{3}\right)}{4\sqrt{2}}$	$2^{-\frac{1}{2}+\frac{3\left(1+\sqrt{3}\right)}{4\sqrt{2}}}$
1	$-\frac{1}{2}+\frac{3}{2\sqrt{2}}$	$2^{-\frac{1}{2}+\frac{3}{2\sqrt{2}}}$
2	$-\frac{1}{2}+\frac{3\left(-1+\sqrt{3}\right)}{4\sqrt{2}}$	$2^{-\frac{1}{2}+\frac{3\left(-1+\sqrt{3}\right)}{4\sqrt{2}}}$
3	$-\frac{1}{2}-\frac{3\left(-1+\sqrt{3}\right)}{4\sqrt{2}}$	$2^{-\frac{1}{2}-\frac{3\left(-1+\sqrt{3}\right)}{4\sqrt{2}}}$
4	$-\frac{1}{2}-\frac{3}{2\sqrt{2}}$	$2^{-\frac{1}{2}-\frac{3}{2\sqrt{2}}}$
5	$-\frac{1}{2}-\frac{3\left(1+\sqrt{3}\right)}{4\sqrt{2}}$	$2^{-\frac{1}{2}-\frac{3\left(1+\sqrt{3}\right)}{4\sqrt{2}}}$

In[49]:=
```
n = 10;
k = 0;
xs10 = {};
fxs10 = {};
While[k ≤ n,
  xk = (a + b) / 2 + (b - a) / 2 * Cos[(2 k + 1) * Pi / (2 (n + 1))];
  xs10 = Append[xs10, xk];
  fxs10 = Append[fxs10, f[xk]];
  k = k + 1;
 ];
ks = Range[0, n];
data = Transpose[{ks, xs10, fxs10}];

TableForm[data, TableHeadings → {None, {"j", "xk", "f(xk)"}}]
```

Out[56]//TableForm=

j	xk	f(xk)
0	$-\frac{1}{2}+\frac{3}{2}\,\text{Cos}\left[\frac{\pi}{22}\right]$	$2^{-\frac{1}{2}+\frac{3}{2}\text{Cos}\left[\frac{\pi}{22}\right]}$
1	$-\frac{1}{2}+\frac{3}{2}\,\text{Cos}\left[\frac{3\pi}{22}\right]$	$2^{-\frac{1}{2}+\frac{3}{2}\text{Cos}\left[\frac{3\pi}{22}\right]}$
2	$-\frac{1}{2}+\frac{3}{2}\,\text{Cos}\left[\frac{5\pi}{22}\right]$	$2^{-\frac{1}{2}+\frac{3}{2}\text{Cos}\left[\frac{5\pi}{22}\right]}$
3	$-\frac{1}{2}+\frac{3}{2}\,\text{Sin}\left[\frac{2\pi}{11}\right]$	$2^{-\frac{1}{2}+\frac{3}{2}\text{Sin}\left[\frac{2\pi}{11}\right]}$
4	$-\frac{1}{2}+\frac{3}{2}\,\text{Sin}\left[\frac{\pi}{11}\right]$	$2^{-\frac{1}{2}+\frac{3}{2}\text{Sin}\left[\frac{\pi}{11}\right]}$
5	$-\frac{1}{2}$	$\frac{1}{\sqrt{2}}$
6	$-\frac{1}{2}-\frac{3}{2}\,\text{Sin}\left[\frac{\pi}{11}\right]$	$2^{-\frac{1}{2}-\frac{3}{2}\text{Sin}\left[\frac{\pi}{11}\right]}$
7	$-\frac{1}{2}-\frac{3}{2}\,\text{Sin}\left[\frac{2\pi}{11}\right]$	$2^{-\frac{1}{2}-\frac{3}{2}\text{Sin}\left[\frac{2\pi}{11}\right]}$
8	$-\frac{1}{2}-\frac{3}{2}\,\text{Cos}\left[\frac{5\pi}{22}\right]$	$2^{-\frac{1}{2}-\frac{3}{2}\text{Cos}\left[\frac{5\pi}{22}\right]}$
9	$-\frac{1}{2}-\frac{3}{2}\,\text{Cos}\left[\frac{3\pi}{22}\right]$	$2^{-\frac{1}{2}-\frac{3}{2}\text{Cos}\left[\frac{3\pi}{22}\right]}$
10	$-\frac{1}{2}-\frac{3}{2}\,\text{Cos}\left[\frac{\pi}{22}\right]$	$2^{-\frac{1}{2}-\frac{3}{2}\text{Cos}\left[\frac{\pi}{22}\right]}$

```
*)
data1 = Transpose[{xs1, fxs1}]
P1 = InterpolatingPolynomial[data1, x]
```

Out[83]= $\left\{\left\{-\frac{1}{2}+\frac{3}{2\sqrt{2}}, 2^{-\frac{1}{2}+\frac{3}{2\sqrt{2}}}\right\}, \left\{-\frac{1}{2}-\frac{3}{2\sqrt{2}}, 2^{-\frac{1}{2}-\frac{3}{2\sqrt{2}}}\right\}\right\}$

Out[84]= $2^{-\frac{1}{2}+\frac{3}{2\sqrt{2}}} - \frac{1}{3}\sqrt{2}\left(2^{-\frac{1}{2}-\frac{3}{2\sqrt{2}}} - 2^{-\frac{1}{2}+\frac{3}{2\sqrt{2}}}\right)\left(\frac{1}{2}-\frac{3}{2\sqrt{2}}+x\right)$

plot | simplify | x derivative | x integral | more...

In[67]:=
```
data2 = Transpose[{xs2, fxs2}]
P2 = Expand[N[InterpolatingPolynomial[data2, x]]]
```

Out[67]= $\left\{\left\{-\frac{1}{2}+\frac{3\sqrt{3}}{4}, 2^{-\frac{1}{2}+\frac{3\sqrt{3}}{4}}\right\}, \left\{-\frac{1}{2}, \frac{1}{\sqrt{2}}\right\}, \left\{-\frac{1}{2}-\frac{3\sqrt{3}}{4}, 2^{-\frac{1}{2}-\frac{3\sqrt{3}}{4}}\right\}\right\}$

Out[68]= $1.03207 + 0.740753\,x + 0.181657\,x^2$

In[79]:=
```
data3 = Transpose[{N[xs3], N[fxs3]}]
P3 = Expand[N[InterpolatingPolynomial[data3, x]]]
```

Out[79]= {{0.885819, 1.84781}, {0.0740251, 1.05265}, {-1.07403, 0.474992}, {-1.88582, 0.27059}}

Out[80]= $0.998996 + 0.706238\,x + 0.247784\,x^2 + 0.0414171\,x^3$

In[81]:=
```
data4 = Transpose[{N[xs4], N[fxs4]}]
P4 = Expand[N[InterpolatingPolynomial[data4, x]]]
```

Out[81]= {{0.926585, 1.90077}, {0.381678, 1.30286}, {-0.5, 0.707107}, {-1.38168, 0.383772}, {-1.92658, 0.263051}}

Out[82]= $0.999514 + 0.693311\,x + 0.243312\,x^2 + 0.056195\,x^3 + 0.00711336\,x^4$

In[75]:=
```
data5 = Transpose[{N[xs5], N[fxs5]}]
P5 = Expand[N[InterpolatingPolynomial[data5, x]]]
```

Out[75]= {{0.948889, 1.93039}, {0.56066, 1.47494}, {-0.111771, 0.925451}, {-0.888229, 0.540277}, {-1.56066, 0.338996}, {-1.94889, 0.259016}}

Out[76]= $0.999981 + 0.692986\,x + 0.240361\,x^2 + 0.056019\,x^3 + 0.00962634\,x^4 + 0.000979846\,x^5$

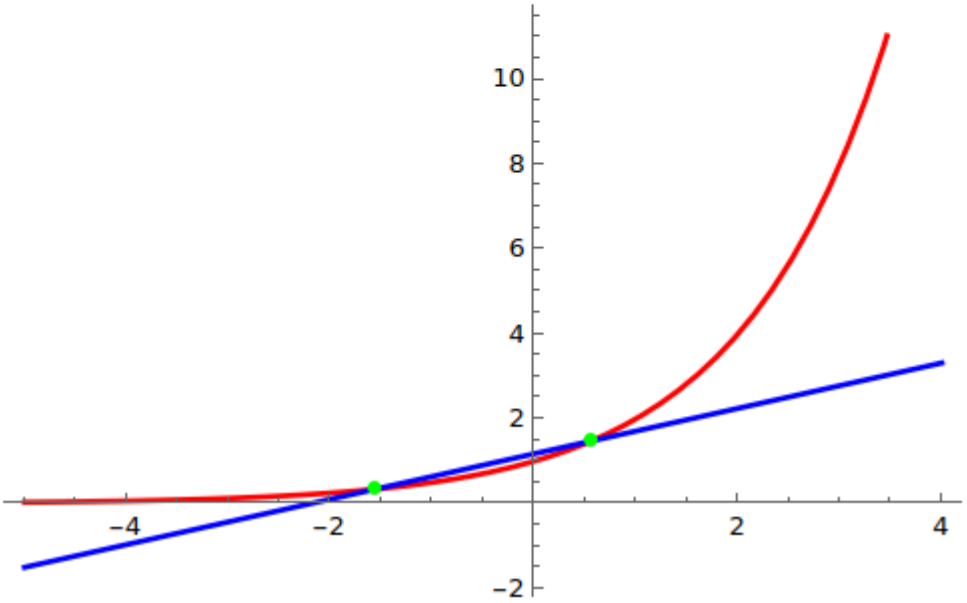
In[77]:=
```
data10 = Transpose[{N[xs10], N[fxs10]}]
P10 = Expand[N[InterpolatingPolynomial[data10, x]]]
```

Out[77]= {{0.984732, 1.97895}, {0.864448, 1.82064}, {0.633624, 1.55146}, {0.310961, 1.24053}, {-0.0774012, 0.947763}, {-0.5, 0.707107}, {-0.922599, 0.527558}, {-1.31096, 0.403052}, {-1.63362, 0.322278}, {-1.86445, 0.274628}, {-1.98473, 0.25266}}

Out[78]= $1. + 0.693147\,x + 0.240227\,x^2 + 0.0555041\,x^3 + 0.00961813\,x^4 + 0.00133335\,x^5 + 0.000154036\,x^6 + 0.0000152597\,x^7 + 1.32434\times10^{-6}\,x^8 + 9.94492\times10^{-8}\,x^9 + 5.1021\times10^{-9}\,x^{10}$

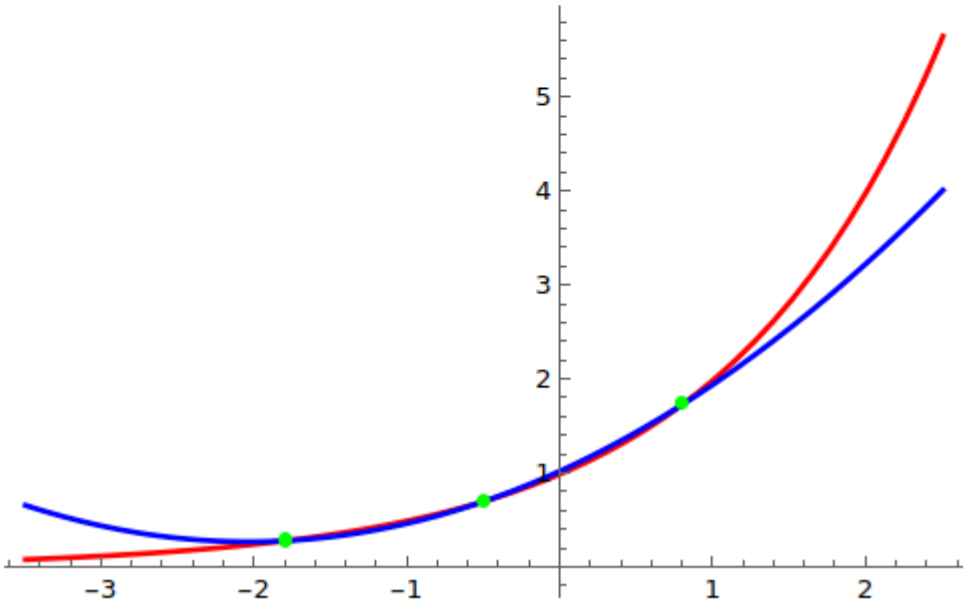
```
n = 1;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P1, {x, xmin, xmax}, PlotStyle → Blue];
p31 = ListPlot[data1, PlotStyle → Green];
Show[p1, p2, p31, PlotRange → All]
```

Out[92]=



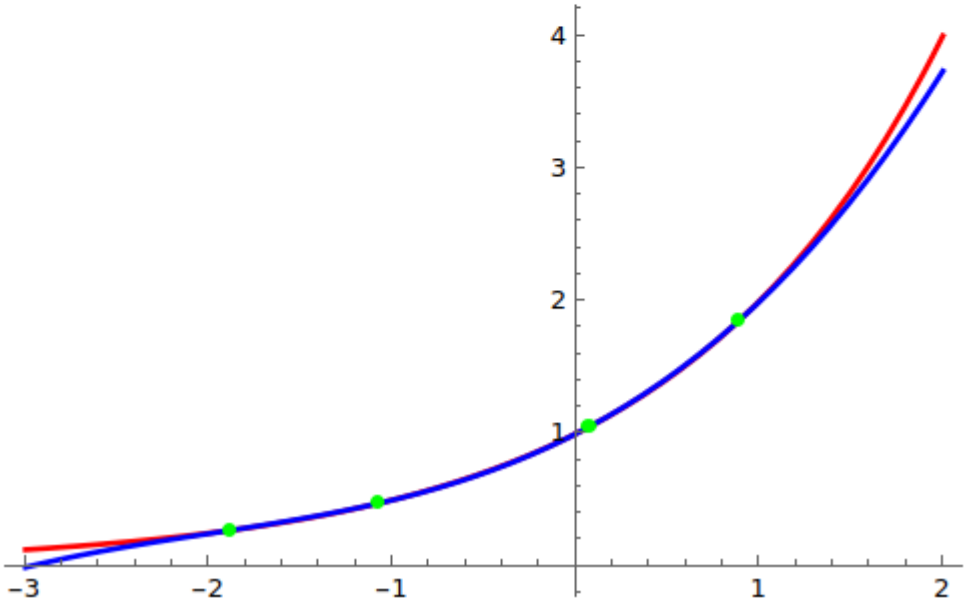
In[93]:=
```
n = 2;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P2, {x, xmin, xmax}, PlotStyle → Blue];
p32 = ListPlot[data2, PlotStyle → Green];
Show[p1, p2, p32, PlotRange → All]
```

Out[100]=



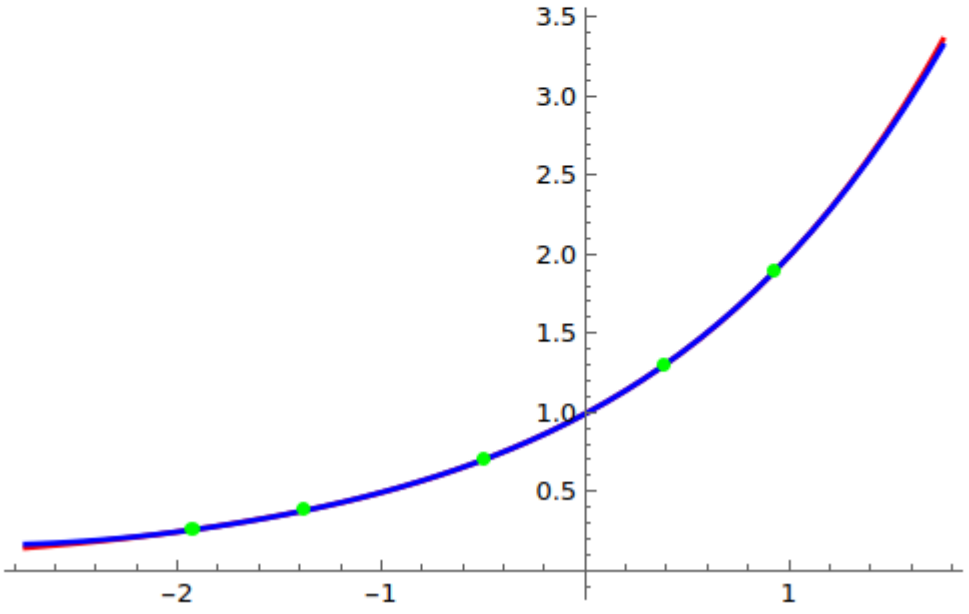
In[101]:=
```
n = 3;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P3, {x, xmin, xmax}, PlotStyle → Blue];
p33 = ListPlot[data3, PlotStyle → Green];
Show[p1, p2, p33, PlotRange → All]
```

Out[108]=



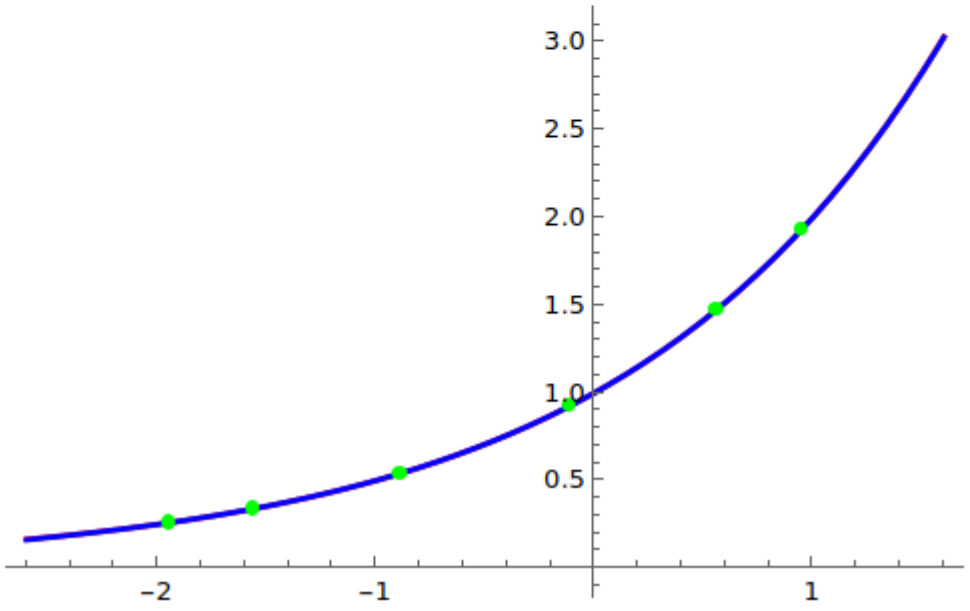
In[109]:=
```
n = 4;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P4, {x, xmin, xmax}, PlotStyle → Blue];
p34 = ListPlot[data4, PlotStyle → Green];
Show[p1, p2, p34, PlotRange → All]
```

Out[116]=



In[117]:=
```
n = 5;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P5, {x, xmin, xmax}, PlotStyle → Blue];
p35 = ListPlot[data5, PlotStyle → Green];
Show[p1, p2, p35, PlotRange → All]
```

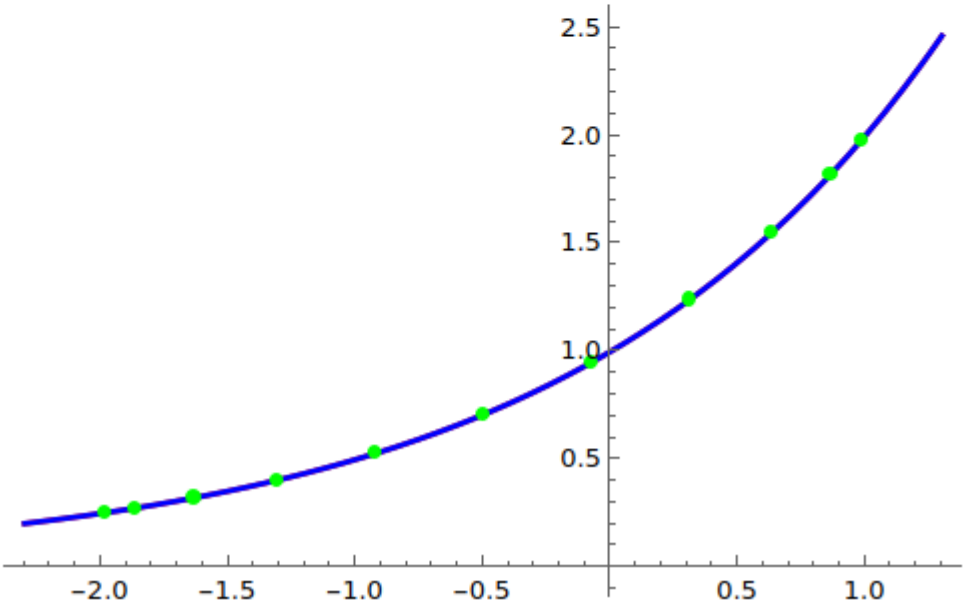
Out[124]=



In[132]:=
```
n = 10;
h = (b - a) / n;
xmin = a - h;

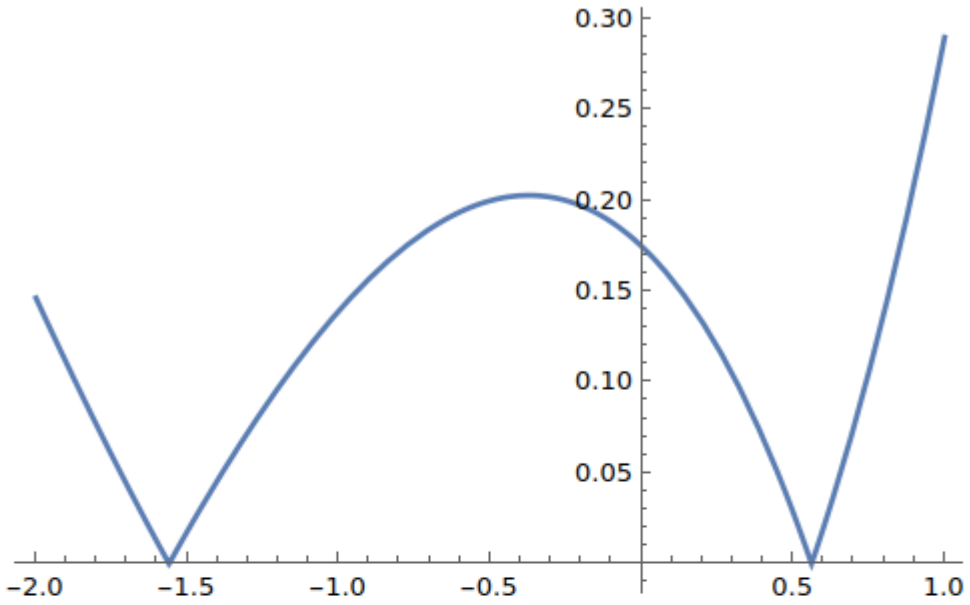
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P10, {x, xmin, xmax}, PlotStyle → Blue];
p310 = ListPlot[data10, PlotStyle → Green];
Show[p1, p2, p310, PlotRange → All]
```

Out[139]=



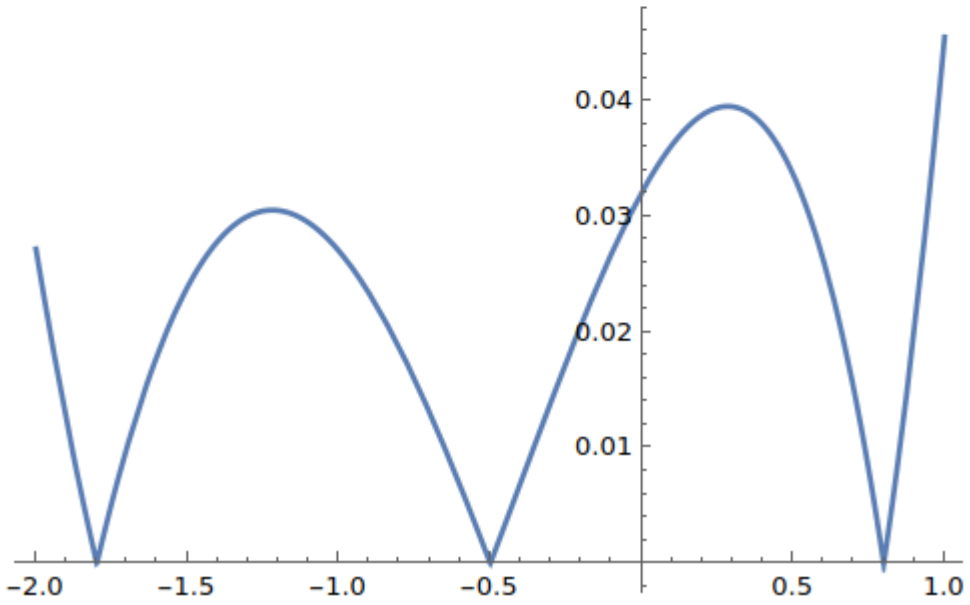
In[140]:=
```
Rn1 = Plot[Abs[f[x] - P1], {x, a, b}, PlotRange → All]
```

Out[140]=



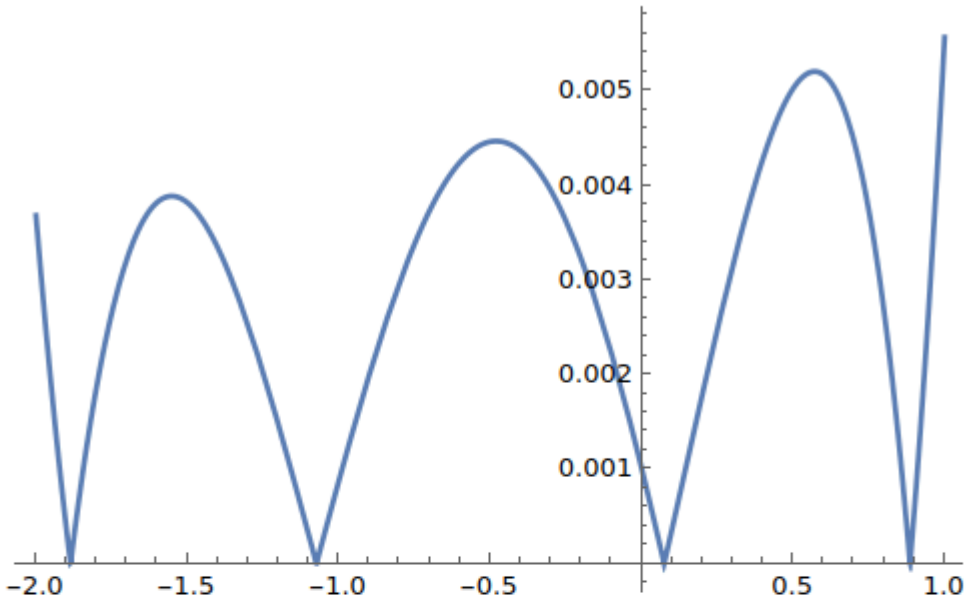
In[141]:=
```
Rn2 = Plot[Abs[f[x] - P2], {x, a, b}, PlotRange → All]
```

Out[141]=



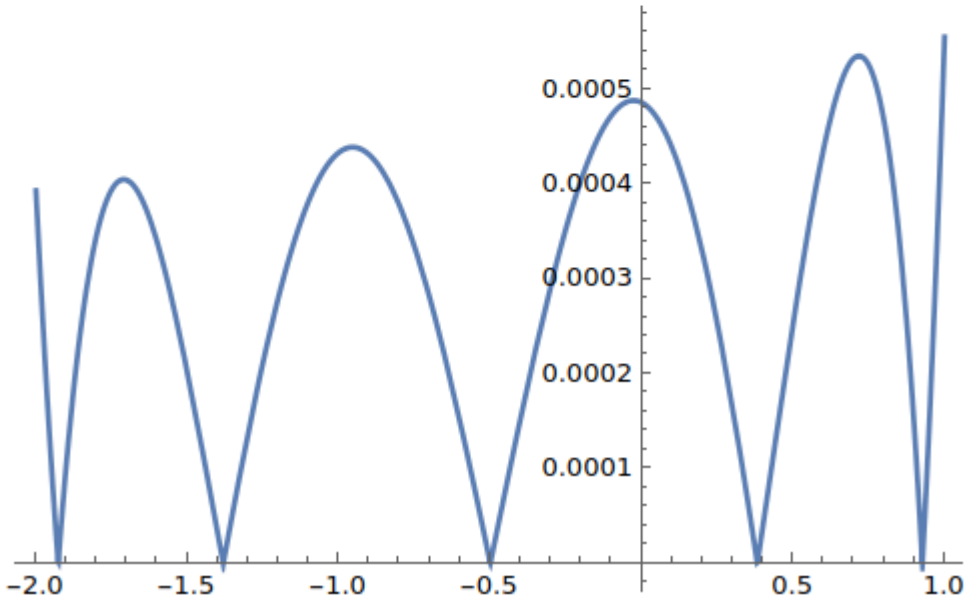
In[142]:=
```
Rn3 = Plot[Abs[f[x] - P3], {x, a, b}, PlotRange → All]
```

Out[142]=



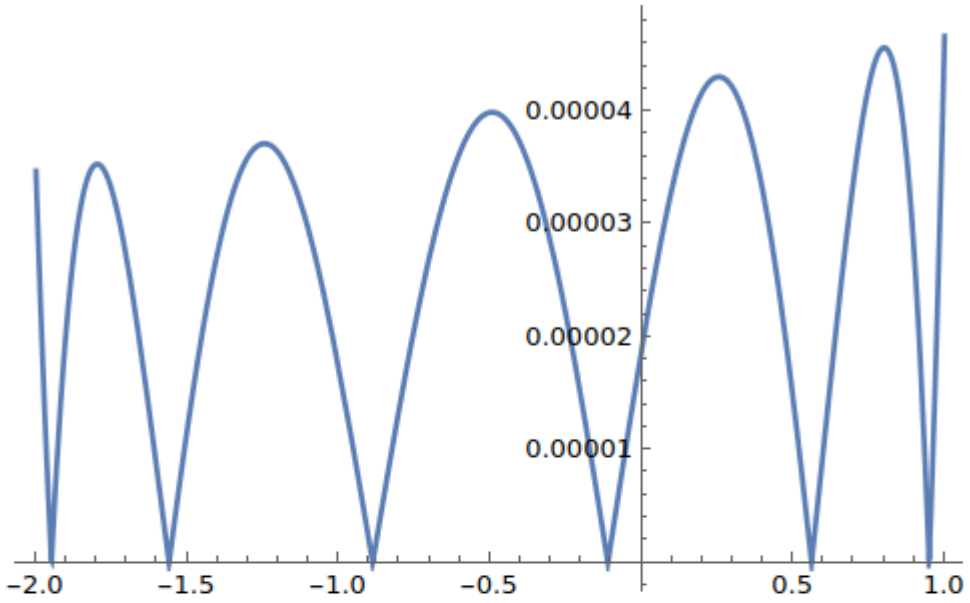
In[143]:=
```
Rn4 = Plot[Abs[f[x] - P4], {x, a, b}, PlotRange → All]
```

Out[143]=



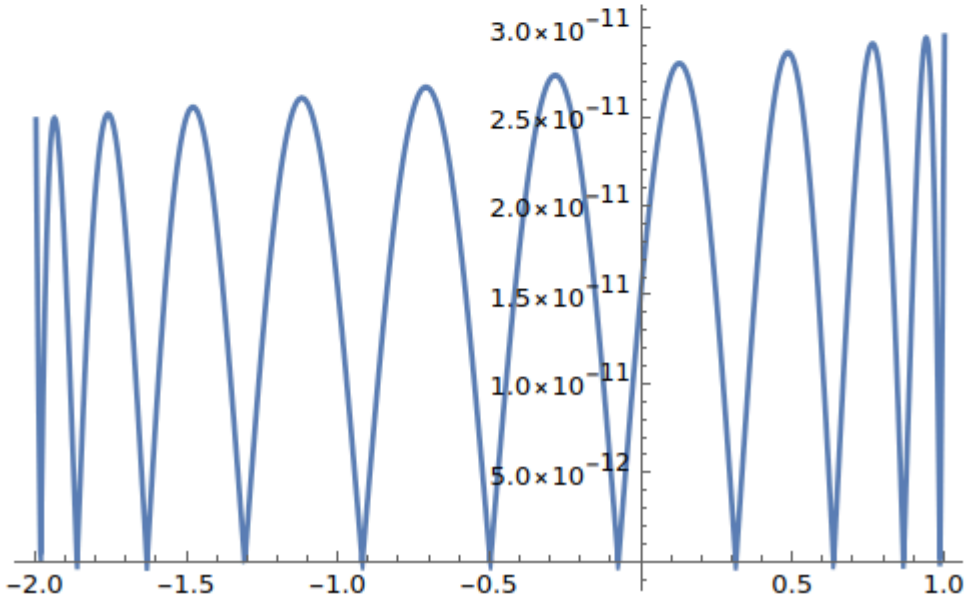
In[144]:=
```
Rn5 = Plot[Abs[f[x] - P5], {x, a, b}, PlotRange → All]
```

Out[144]=



In[145]:=
```
Rn10 = Plot[Abs[f[x] - P10], {x, a, b}, PlotRange → All
```

Out[145]=



In[152]:= **FindMaximum[{Abs[f[x] - P1], a ≤ x ≤ b}, x]**

Out[152]= {0.289793, {x → 1.}}

In[147]:= **FindMaximum[{Abs[f[x] - P2], a ≤ x ≤ b}, x]**

Out[147]= {0.0395289, {x → 0.28392}}

In[148]:= **FindMaximum[{Abs[f[x] - P3], a ≤ x ≤ b}, x]**

Out[148]= {0.00520315, {x → 0.571158}}

In[149]:= **FindMaximum[{Abs[f[x] - P4], a ≤ x ≤ b}, x]**

Out[149]= {0.000535053, {x → 0.717419}}

In[150]:= **FindMaximum[{Abs[f[x] - P5], a ≤ x ≤ b}, x]**

Out[150]= {0.0000465681, {x → 0.99992}}

In[151]:= **FindMaximum[{Abs[f[x] - P10], a ≤ x ≤ b}, x]**

Out[151]= $\{2.84107 \times 10^{-11}, \{x \to 0.468745\}\}$

```
(*
 3. Выполните интерполяцию функции f (x) сплайном Sf (x) с помощью
функции Interpolation[data,Method→“Spline”],используя таблицу
  значений функции f (x) в равноотстоящих точках для n=5 и 10:
*)
```

```
n = 5;
j = 0;
xs5 = {};
fxs5 = {};
While[j ≤ n,
  xj = a + (b - a) / n * j;
  xs5 = Append[xs5, xj];
  fxs5 = Append[fxs5, f[xj]];
  j = j + 1;
 ];
js = Range[0, n];
data = Transpose[{js, xs5, fxs5}];
TableForm[data, TableHeadings → {None, {"j", "xj", "f(xj)"}}]
```

Out[168]//TableForm=

j	xj	f(xj)
0	-2	$\frac{1}{4}$
1	$-\frac{7}{5}$	$\frac{1}{2\times 2^{2/5}}$
2	$-\frac{4}{5}$	$\frac{1}{2^{4/5}}$
3	$-\frac{1}{5}$	$\frac{1}{2^{1/5}}$
4	$\frac{2}{5}$	$2^{2/5}$
5	1	2

In[•]:=

```
n = 10;
j = 0;
xs10 = {};
fxs10 = {};
While[j ≤ n,
  xj = a + (b - a) / n * j;
  xs10 = Append[xs10, xj];
  fxs10 = Append[fxs10, f[xj]];
  j = j + 1;
 ];
js = Range[0, n];
data = Transpose[{js, xs10, fxs10}];

TableForm[data, TableHeadings → {None, {"j", "xj", "f(xj)"}}]
```

Out[•]//TableForm=

j	xj	f(xj)
0	-2	$\frac{1}{4}$
1	$-\frac{17}{10}$	$\frac{1}{2\times 2^{7/10}}$
2	$-\frac{7}{5}$	$\frac{1}{2\times 2^{2/5}}$
3	$-\frac{11}{10}$	$\frac{1}{2\times 2^{1/10}}$
4	$-\frac{4}{5}$	$\frac{1}{2^{4/5}}$
5	$-\frac{1}{2}$	$\frac{1}{\sqrt{2}}$
6	$-\frac{1}{5}$	$\frac{1}{2^{1/5}}$
7	$\frac{1}{10}$	$2^{1/10}$
8	$\frac{2}{5}$	$2^{2/5}$
9	$\frac{7}{10}$	$2^{7/10}$
10	1	2

```
data5 = Transpose[{N[xs5], N[fxs5]}]
P5 = Interpolation[data5, Method → "Spline"]
```

Out[193]= {{-2., 0.25}, {-1.4, 0.378929},
{-0.8, 0.574349}, {-0.2, 0.870551}, {0.4, 1.31951}, {1., 2.}}

Out[194]= InterpolatingFunction[ Domain: {{-2., 1.}}
Output: scalar]

evaluate at point... | differentiate | integrate

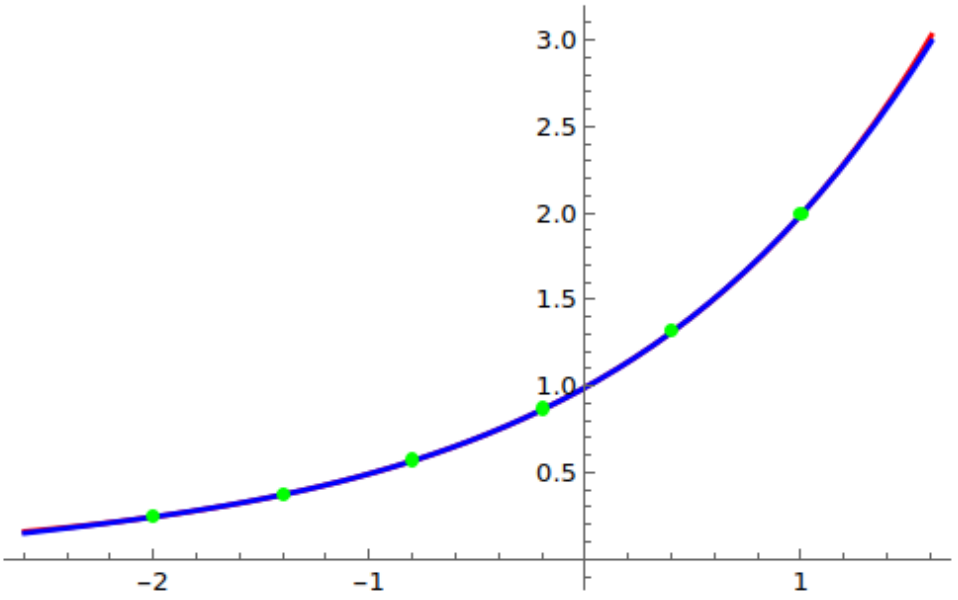
In[191]:=
```
data10 = Transpose[{N[xs10], N[fxs10]}]
P10 = Interpolation[data10, Method → "Spline"]
```

Out[191]= {{-2., 0.25}, {-1.7, 0.307786}, {-1.4, 0.378929},
{-1.1, 0.466516}, {-0.8, 0.574349}, {-0.5, 0.707107}, {-0.2, 0.870551},
{0.1, 1.07177}, {0.4, 1.31951}, {0.7, 1.6245}, {1., 2.}}

Out[192]= InterpolatingFunction[ Domain: {{-2., 1.}}
Output: scalar]

In[203]:=
```
n = 5;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P5[x], {x, xmin, xmax}, PlotStyle → Blue];
p35 = ListPlot[data5, PlotStyle → Green];
Show[p1, p2, p35, PlotRange → All]
```

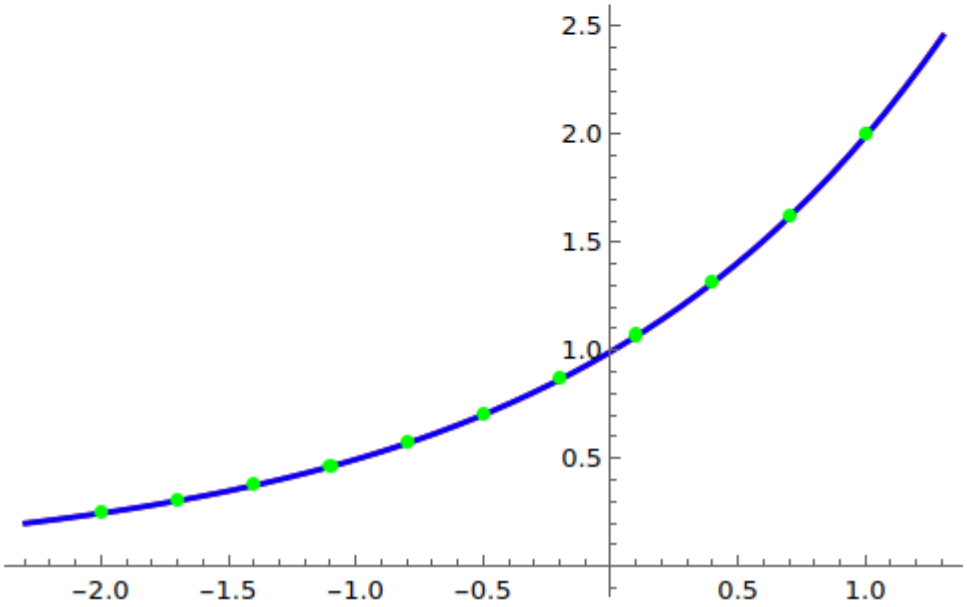
Out[210]=



In[211]:=
```
n = 10;
h = (b - a) / n;
xmin = a - h;

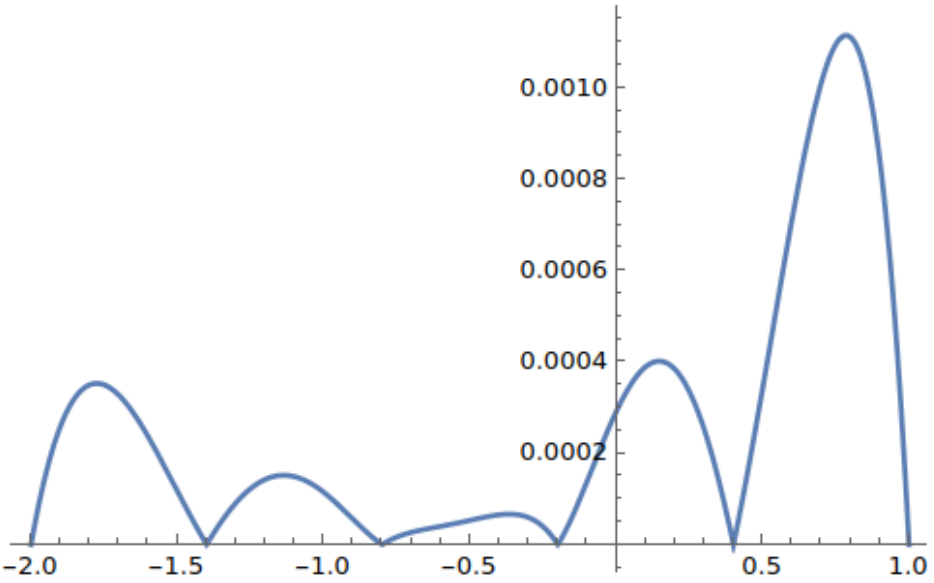
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[P10[x], {x, xmin, xmax}, PlotStyle → Blue];
p310 = ListPlot[data10, PlotStyle → Green];
Show[p1, p2, p310, PlotRange → All]
```

Out[218]=



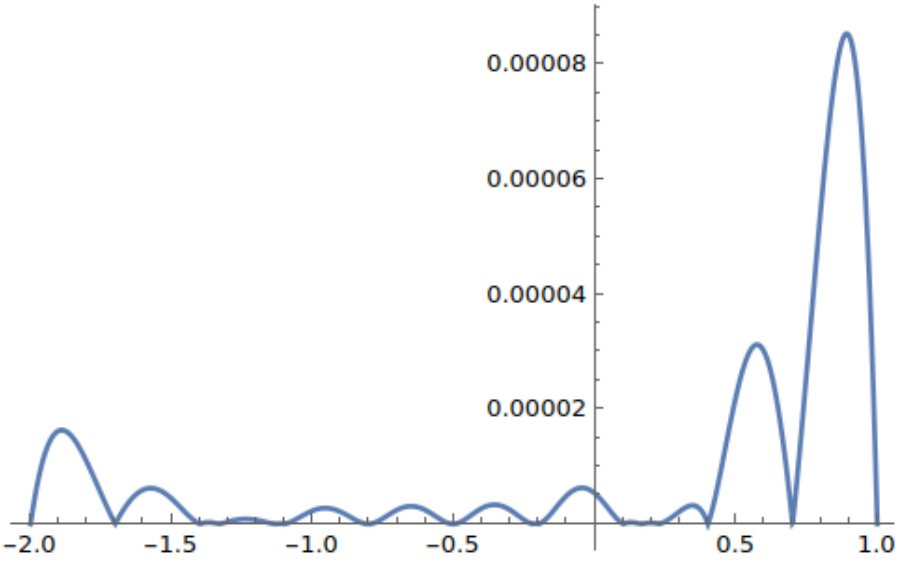
In[219]:=
```
Rn5 = Plot[Abs[f[x] - P5[x]], {x, a, b}, PlotRange → All]
```

Out[219]=



In[220]:=
```
Rn10 = Plot[Abs[f[x] - P10[x]], {x, a, b}, PlotRange → All]
```

Out[220]=



In[221]:=
```
FindMaximum[{Abs[f[x] - P5[x]], a ≤ x ≤ b}, x]
```

Out[221]=
$$\{0.00111462, \{x \to 0.785044\}\}$$

In[222]:=
```
FindMaximum[{Abs[f[x] - P10[x]], a ≤ x ≤ b}, x]
```

Out[222]=
$$\left\{2.25185 \times 10^{-7}, \{x \to 0.198851\}\right\}$$

In[229]:=
```
(*
4. Составьте таблицу максимальных значений погрешности Rn (x) для
 разных значений n по результатам заданий 1,2 и 3. Сравните для
 равноотстоящих и неравноотстоящих узлов и сделайте выводы о зависимость
 погрешности интерполирования от числа узлов и их расположения на
 отрезке.Сравните с результатами интерполирования сплайнами.
*)
dataTask4 = {{1, 0.429937, 0.289793, "-"}, {2, 0.0630579, 0.0395289, "-"},
   {3, 0.00832901, 0.00520315, "-"}, {4, 0.000974132, 0.000535053, "-"},
   {5, 0.000100931, 0.0000465681, 0.00111462},
   {10, 2.43027 * 10^-11, 2.84107 * 10^-11, 2.25185 * 10^-7}};

headers = {"n", "1 задание", "2 задание (чебышев)", "3 задание (сплайн"};

TableForm[dataTask4, TableHeadings → {None, headers}, TableAlignments → Center]
```

Out[231]//TableForm=

n	1 задание	2 задание (чебышев)	3 задание (сплайн
1	0.429937	0.289793	-
2	0.0630579	0.0395289	-
3	0.00832901	0.00520315	-
4	0.000974132	0.000535053	-
5	0.000100931	0.0000465681	0.00111462
10	2.43027×10^{-11}	2.84107×10^{-11}	2.25185×10^{-7}

```
(*
Вывод: с увеличением числа узлов погрешность падает и соответственно
  возрастает точность. Равномерное распределение проигрывает
  распределению чебышева в точности во всех узлах кроме 10, тк
 в оьоих случаях там достигнута крайне высокая точность.
  Сплайн проигрывает обоим случаем в точности для заданной функции.
  \*)
```

```
(*
   5. Постройте с помощью метода наименьших квадратов
     аппроксимирующие алгебраические многочлены наилучшего среднеквадратичного
     приближения.Используйте функции Fit и FindFit пакета Mathematica и
таблицу значений функции f (x) в равноотстоящих точках для n=5 и 10:
     а) постройте многочлены наилучшего среднеквадратичного приближения
степеней m=1,2,3 и 4-Q1 (x),Q2 (x),Q3 (x) и Q4 (x);
*)
Q1 = Fit[data5, {1, x}, x]
FindFit[data5, c + p * x, {c, p}, x]
```

Out[243]= $1.18146 + 0.56514\,x$

Out[244]= $\{c \to 1.18146, p \to 0.56514\}$

In[249]:=
```
Q2 = Fit[data5, {1, x, x^2}, x]
FindFit[data5, a2 + b2 * x + c2 * x^2, {a2, b2, c2}, x]
```

Out[249]= $1.03178 + 0.752241\,x + 0.187101\,x^2$

Out[250]= $\{a2 \to 1.03178, b2 \to 0.752241, c2 \to 0.187101\}$

In[251]:=
```
Q3 = Fit[data5, {1, x, x^2, x^3}, x]
FindFit[data5, a3 + b3 * x + c3 * x^2 + d3 * x^3, {a3, b3, c3, d3}, x]
```

Out[251]= $0.998804 + 0.707323\,x + 0.250189\,x^2 + 0.0420586\,x^3$

Out[252]= $\{a3 \to 0.998804, b3 \to 0.707323, c3 \to 0.250189, d3 \to 0.0420586\}$

In[253]:=
```
Q4 = Fit[data5, {1, x, x^2, x^3, x^4}, x]
FindFit[data5, a4 + b4 * x + c4 * x^2 + d4 * x^3 + e4 * x^4, {a4, b4, c4, d4, e4}, x]
```

Out[253]= $0.999576 + 0.693428\,x + 0.243446\,x^2 + 0.0563617\,x^3 + 0.00715154\,x^4$

In[328]:=
```
Q110 = Fit[data10, {1, x}, x]
FindFit[data10, c + p * x, {c, p}, x]
```

Out[328]= $1.14805 + 0.555919\ x$

Out[329]= $\{c \to 1.14805,\ p \to 0.555919\}$

In[330]:=
```
Q210 = Fit[data10, {1, x, x^2}, x]
FindFit[data10, a2 + b2 * x + c2 * x^2, {a2, b2, c2}, x]
```

Out[330]= $1.02734 + 0.741627\ x + 0.185708\ x^2$

Out[331]= $\{a2 \to 1.02734,\ b2 \to 0.741627,\ c2 \to 0.185708\}$

In[332]:=
```
Q310 = Fit[data10, {1, x, x^2, x^3}, x]
FindFit[data10, a3 + b3 * x + c3 * x^2 + d3 * x^3, {a3, b3, c3, d3}, x]
```

Out[332]= $0.998943 + 0.705835\ x + 0.248723\ x^2 + 0.04201\ x^3$

Out[333]= $\{a3 \to 0.998943,\ b3 \to 0.705835,\ c3 \to 0.248723,\ d3 \to 0.04201\}$

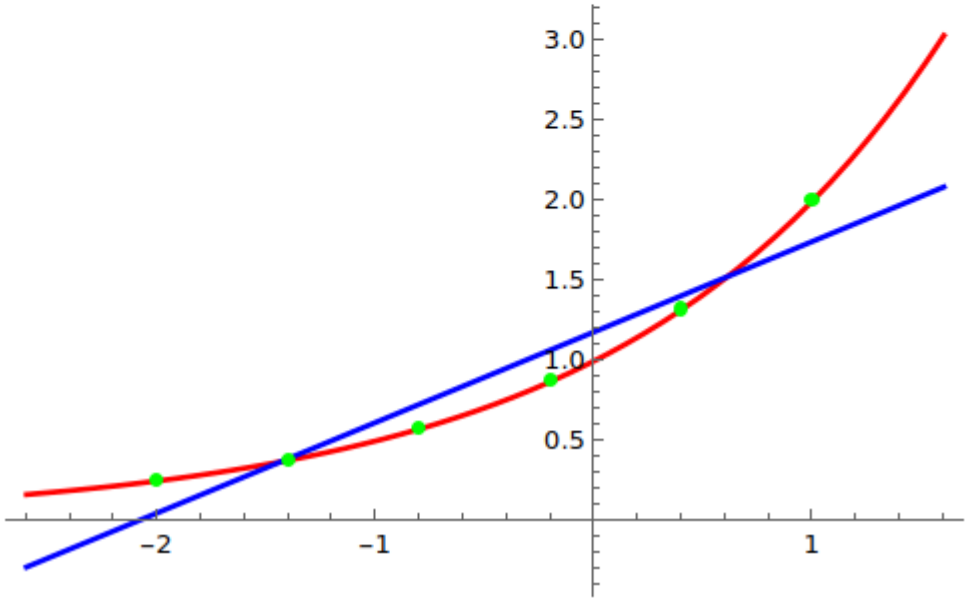
In[334]:=
```
Q410 = Fit[data10, {1, x, x^2, x^3, x^4}, x]
FindFit[data10, a4 + b4 * x + c4 * x^2 + d4 * x^3 + e4 * x^4, {a4, b4, c4, d4, e4}, x]
```

Out[334]= $0.999541 + 0.693271\ x + 0.243339\ x^2 + 0.0563689\ x^3 + 0.00717942\ x^4$

Out[335]= $\{a4 \to 0.999541,\ b4 \to 0.693271,\ c4 \to 0.243339,\ d4 \to 0.0563689,\ e4 \to 0.00717942\}$

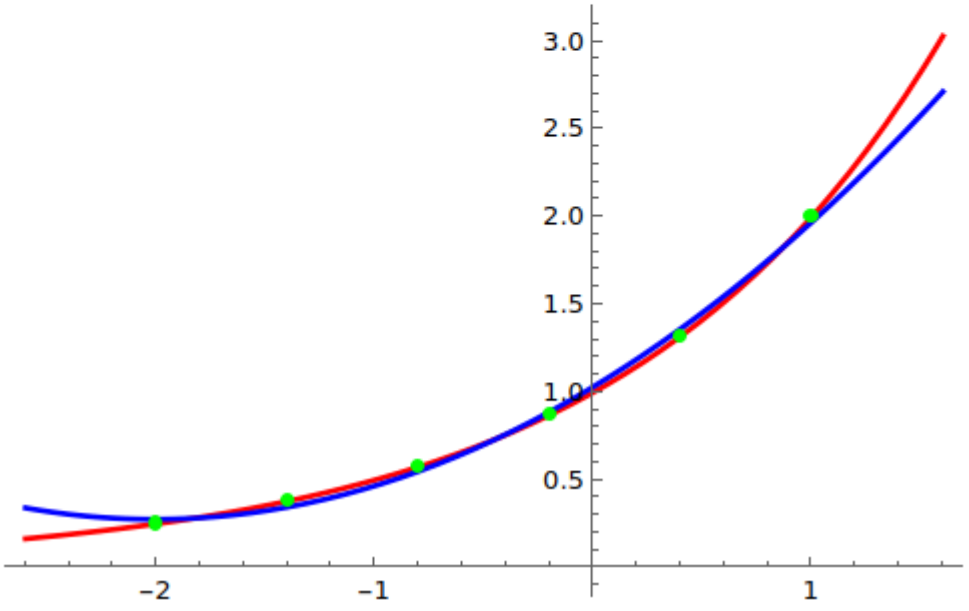
In[344]:=
```
n = 5;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[Q1, {x, xmin, xmax}, PlotStyle → Blue];
p35 = ListPlot[data5, PlotStyle → Green];
Show[p1, p2, p35, PlotRange → All]
```

Out[351]=



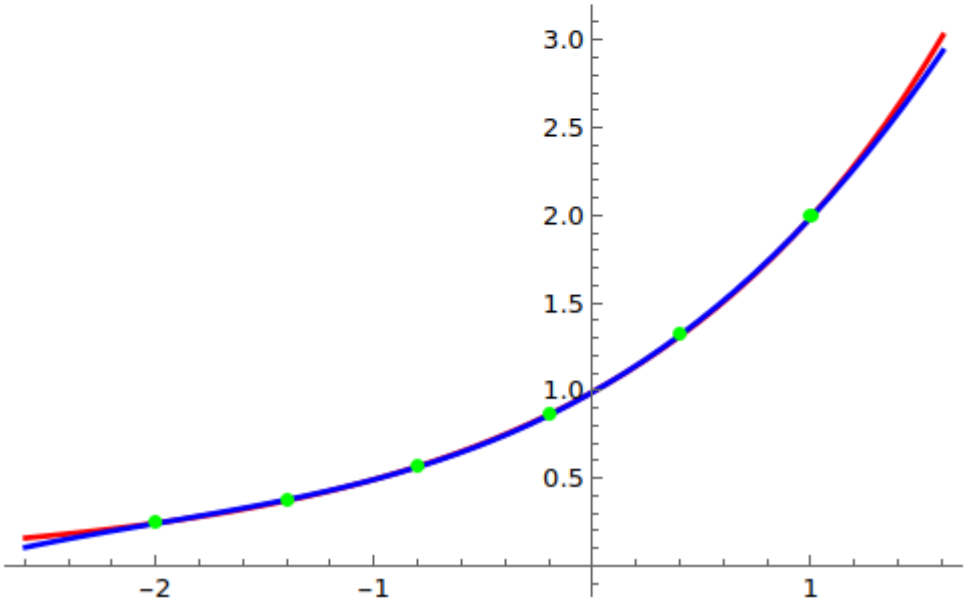
In[352]:=
```
n = 5;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[Q2, {x, xmin, xmax}, PlotStyle → Blue];
p35 = ListPlot[data5, PlotStyle → Green];
Show[p1, p2, p35, PlotRange → All]
```

Out[359]=



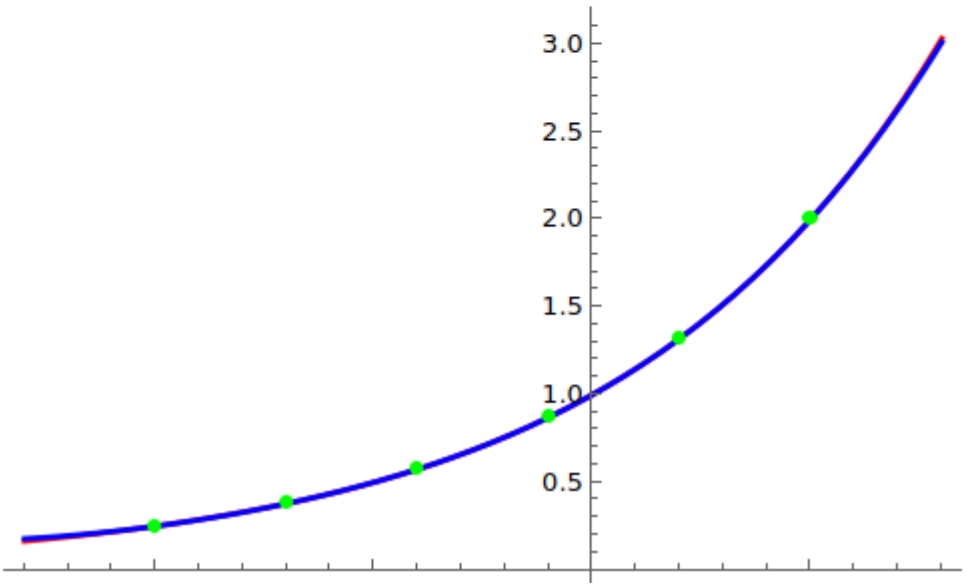
In[360]:=
```
n = 5;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[Q3, {x, xmin, xmax}, PlotStyle → Blue];
p35 = ListPlot[data5, PlotStyle → Green];
Show[p1, p2, p35, PlotRange → All]
```

Out[367]=



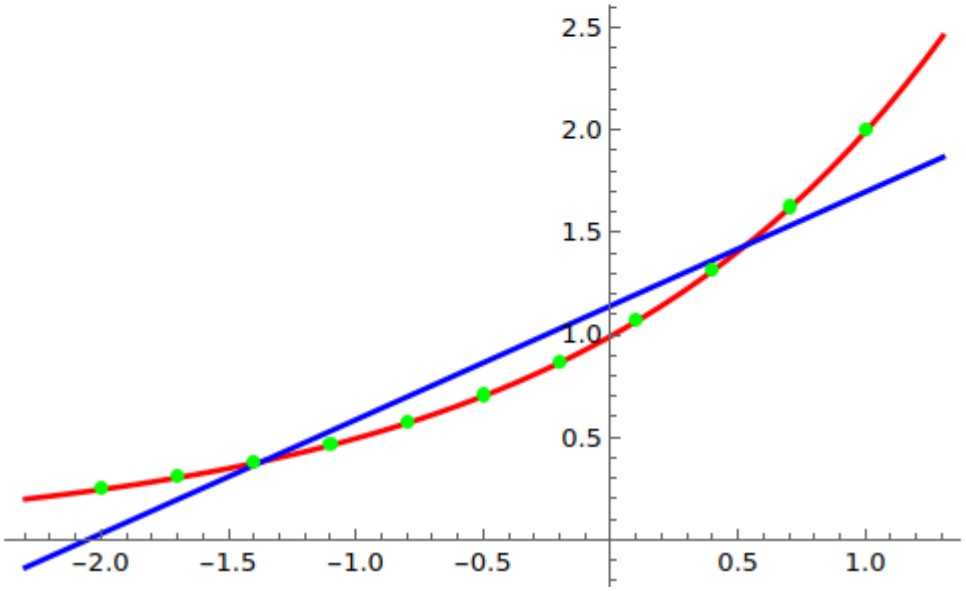
In[368]:=
```
n = 5;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[Q4, {x, xmin, xmax}, PlotStyle → Blue];
p35 = ListPlot[data5, PlotStyle → Green];
Show[p1, p2, p35, PlotRange → All]
```

Out[375]=



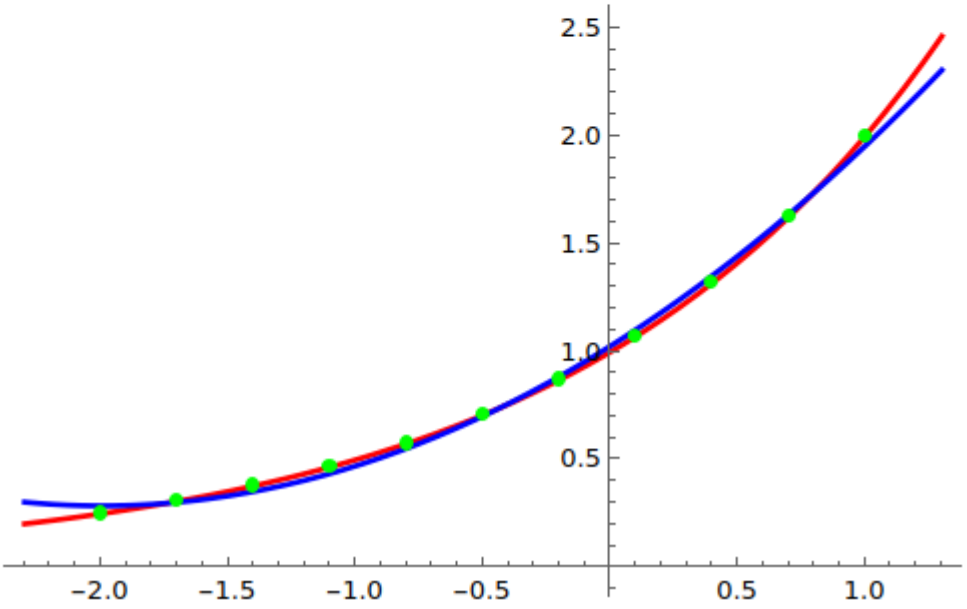
In[384]:=
```
n = 10;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[Q110, {x, xmin, xmax}, PlotStyle → Blue];
p35 = ListPlot[data10, PlotStyle → Green];
Show[p1, p2, p35, PlotRange → All]
```

Out[391]=



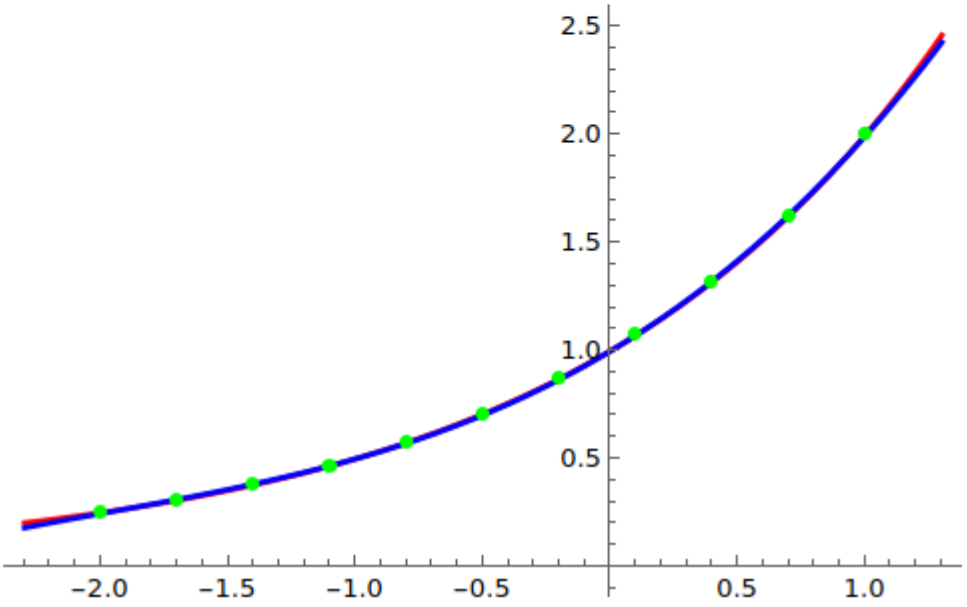
In[392]:=
```
n = 10;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[Q210, {x, xmin, xmax}, PlotStyle → Blue];
p35 = ListPlot[data10, PlotStyle → Green];
Show[p1, p2, p35, PlotRange → All]
```

Out[399]=



In[400]:=
```
n = 10;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[Q310, {x, xmin, xmax}, PlotStyle → Blue];
p35 = ListPlot[data10, PlotStyle → Green];
Show[p1, p2, p35, PlotRange → All]
```

Out[407]=



In[408]:=
```
n = 10;
h = (b - a) / n;
xmin = a - h;
xmax = b + h;
p1 = Plot[f[x], {x, xmin, xmax}, PlotStyle → Red];
p2 = Plot[Q410, {x, xmin, xmax}, PlotStyle → Blue];
p35 = ListPlot[data10, PlotStyle → Green];
Show[p1, p2, p35, PlotRange → All]
```

Out[415]=

